



BASE24-atm[®]

Olivetti 6000 Device Handler User's Guide

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PREFACE

Intended Audience

This document is for users of the BASE24-atm Olivetti 6000 Device Handler. The intended audience includes personnel responsible for configuring, operating and maintaining the BASE24-atm system.

Users are expected to be familiar with BASE24 functions and device control terminal screens. Users should also consult the relevant BASE24, BASE24-atm, Olivetti or Tandem documentation where appropriate.

This document can be used as an aid to training but readers are advised to attend the relevant training course(s).

This manual documents standard processing for the BASE24-atm Release 5.1 product supporting Olivetti 6000 ATMs.

Location of Information

A list of section headings and related page references are given in the *Contents* page. The headings are function oriented.

Section 1 Overview provides an introduction to the purpose and functionality of the Olivetti 6000 Device Handler.

Section 2—Configuration describes how to configure the Olivetti 6000 ATMs in terms of the LCONF parameters and NEFS declarations.

Section 3 File Access and Maintenance describes how to access details of the Application Files Maintenance (AFM) screens for configuration of the Olivetti 6000 ATMs and online help.

Section 4—Device Control Procedures introduces the general device control procedures applicable to all ATMs, and describes the procedures that are specific to the Olivetti 6000 ATMs.

Section 5—Supported Transactions describes the customer and administrative transaction types supported by the Olivetti 6000 Device Handler.

Section 6 Transaction Security describes the transaction security features of the Olivetti 6000 Device Handler and provides guidance on their use.

Section 7 ATM Printer Support describes how the Olivetti 6000 Device Handler supports receipt printing at the ATM.

Appendix A—Message Types lists the message types involved in device handler/ATM communication.

Appendix B ATM Firmware describes the currently supported ATM firmware versions.

User's Comments Form



Related Documentation

BASE24 Documentation

- BASE24 Base Files Maintenance Manual
Publications Number : BA-AE000-03
- BASE24 Configuration Manual
Publications Number : BMSP-NT000-01
- BASE24 Device Control Manual
Publications Number : BA-AE000-06
- BASE24 Log Message Reference Manual
Publications Number : BA-AE000-07
- BASE24 Logical Network Configuration File Manual
Publications Number : BA-AE000-10
- BASE24 Network Control Operators Manual
Publications Number : BMSP-NT000-03
- BASE24 Network Environment Source File Manual
Publications Number : BMSP-NT000-02
- BASE24 SNAX Support Manual **
Publications Number : BMSP-PL021-01 **
- BASE24 Tokens Manual
Publications Number : BA-AE000-12
- BASE24 Transaction Security Manual
Publications Number : BA-AE000-13
- BASE24 X.25 Support Manual **
Publications Number : BMSP-PL030-01 **

** The documentation is available in the Device Handler/Interface code subvolume to be printed by the customer.

BASE24-atm Documentation

- BASE24-atm Files Maintenance Manual
Publications Number : AT-AE000-02
- BASE24-atm Transaction Processing Manual
Publications Number : AT-AE000-07

Olivetti Documentation

- Communications and Networking on PB Protocol Management Guides (for X.25, BSC3 and SNA)
- Hardware Operating Guides (for ATS6400/E, CD6300 and CAT670x)
- SST6000—Application Offline Tools (AOT) System Engineer's Guide
- SST6000 Attached to Financial Transaction Processing Systems Functional SST Specification—6000 Project
- OLIVETTI 6000/ BASE24 Connection Getting Started/Personalisation Guide (Release 4.0)
- OLIVETTI 6000/BASE24 Connection Operating Guide (Release 4.0)

Tandem Documentation

- SNAX/HLS Manual
- System Procedure Errors and Messages Manual
- Systems Management Manual
- Tandem PATHWAY Operating Manual

User Comments

At the back of this manual, you will find a form that can be used to comment on any aspect of this document.

We hope that you will be able to help by telling us what you like and what improvements you would like to see in future editions.



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1. OVERVIEW

1.1 OLIVETTI 6000 ATM

The Olivetti 6000 series ATMs comprise:

- the Automated Teller Machine (ATM)
- the Customer Activated Terminal (CAT)
- the CD6300 LOBBY machine.

Each 6000 series ATM comprises:

- The customer terminal; that is, the set of peripheral devices, such as the cash dispenser, that make up the ATM.
- An integrated, intelligent controller, based on Olivetti personal computer (PC) technology.
- A configuration file that defines the functionality of a particular terminal.

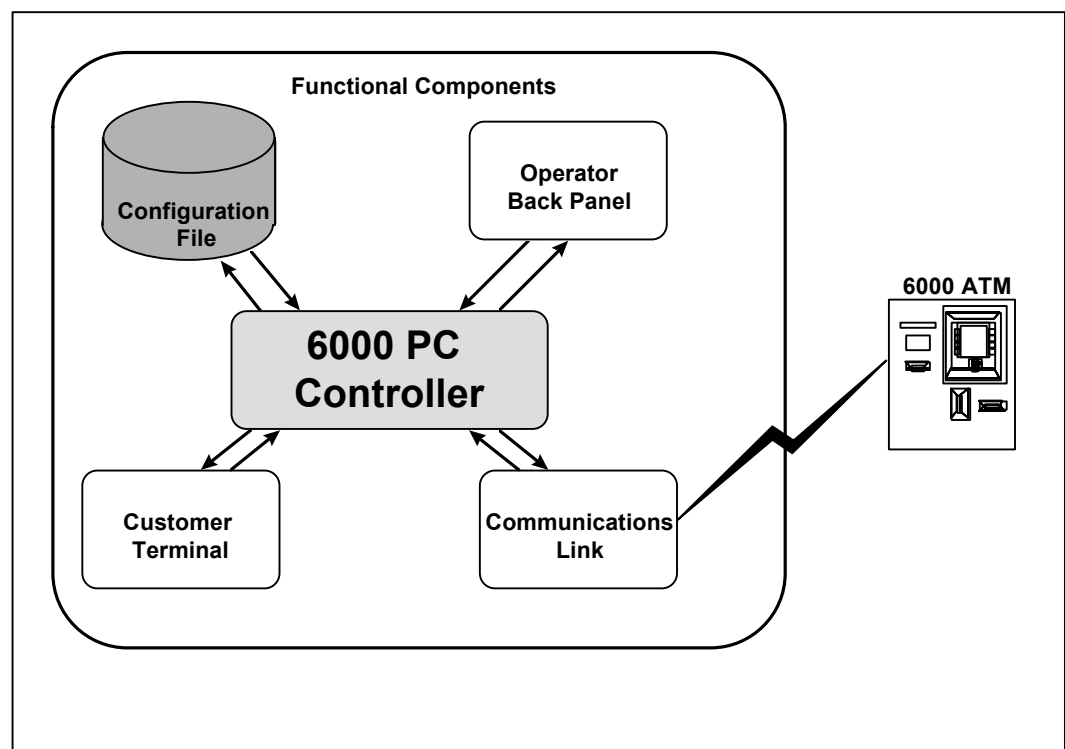


Figure 1-1 The Olivetti 6000 Series ATMs

The use of PC technology for the device controller and its accompanying configuration file, allows an institution great flexibility in defining the functionality of a particular device. It also allows much of the functionality to be controlled locally at the ATM, rather than relying on a remote site for processing requirements.

BASE24 supports communications with Olivetti 6000 ATMs via the bisynchronous, SNA/SDLC, SNAX/HLS, TCP/IP and X.25 protocols.

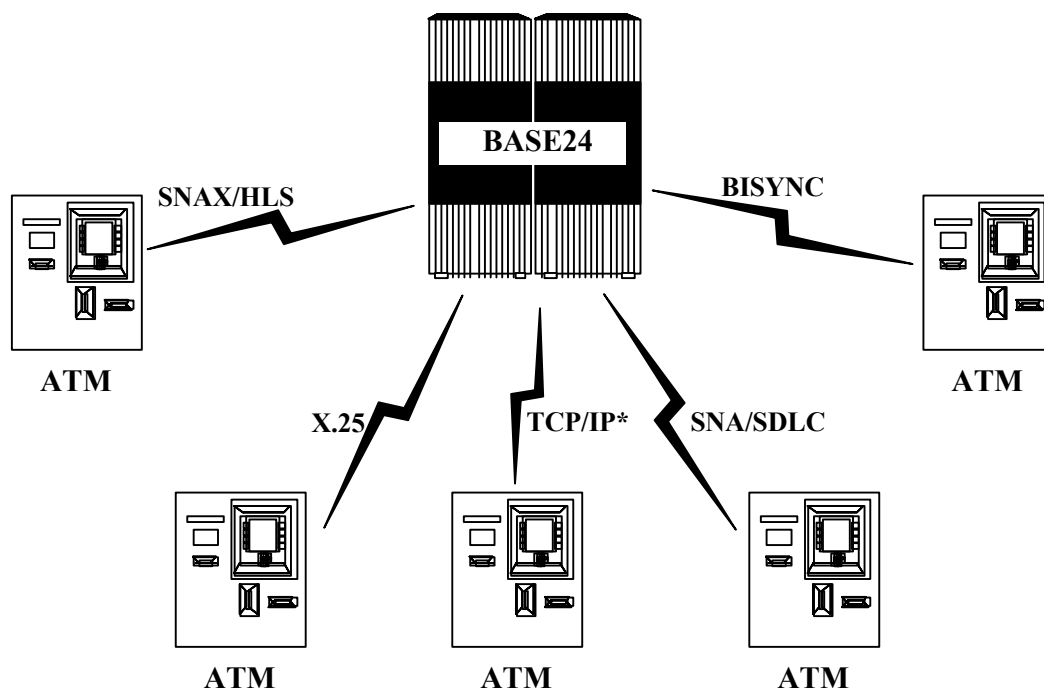


Figure 1-2 ATM/BASE24 Communications

* For TCP/IP support please contact your local Olivetti Office.

1.2 OLIVETTI 6000 DEVICE HANDLER

Olivetti 6000 ATMs are connected to BASE24-atm transaction processing through the Olivetti 6000 Device Handler process. Every transaction initiated at the ATM is passed to this Device Handler process, via XPNET, when it enters the BASE24-atm system. The message coming from the ATM is in the device native mode and the Device Handler converts this message to BASE24 Standard Internal Message (STM) format for sending to a BASE24 Authorisation process. Both customer and administrative transactions are supported.

Transactions processed at different terminals are handled differently according to how an individual terminal is configured. At the network level, the transaction processing for the terminal is configured in the BASE24 Terminal Data File (TDF). Each ATM has a record in the TDF. This allows each ATM to have different sharing arrangements and transaction sets. The Device Handler process reads the TDF whenever it needs information on an ATM.

In addition, each Olivetti 6000 ATM has a locally held configuration file. The ATM configuration file contains the control logic and associated parameters that determine how transaction processing is configured at the ATM level; for example, screen data. ATM configuration files can be uploaded from an ATM to BASE24 by the Device Handler and then downloaded to another ATM. Thus, the ATM configuration file can be modified at one Olivetti ATM and these changes can be distributed to the other Olivetti 6000 ATMs in the network via the BASE24 Device Handler.

The BASE24 Institution Data File (IDF) is another file read by the Device Handler process. The IDF obtains the dates the Device Handler process requires for cutting over its terminals.

The Device Handler process also uses the Logical Network Configuration File (LCONF). The Device Handler reads the LCONF to obtain the names of both the TDF and IDF as well as a number of other file names and processing parameters.

The ATM status can be controlled and monitored, by the central operator, via the BASE24 Device Control Terminal (DCT). The Device Handler creates log messages and sends them to the Tandem Event Management system (EMS) via BASE24 Logger process.

Details of the BASE24 files used for the configuration and maintenance of the Olivetti 6000 ATMs are described in sections 2, 3 and 4 of this guide.

Figure 1-3 illustrates the role of the Device Handler during transaction processing.

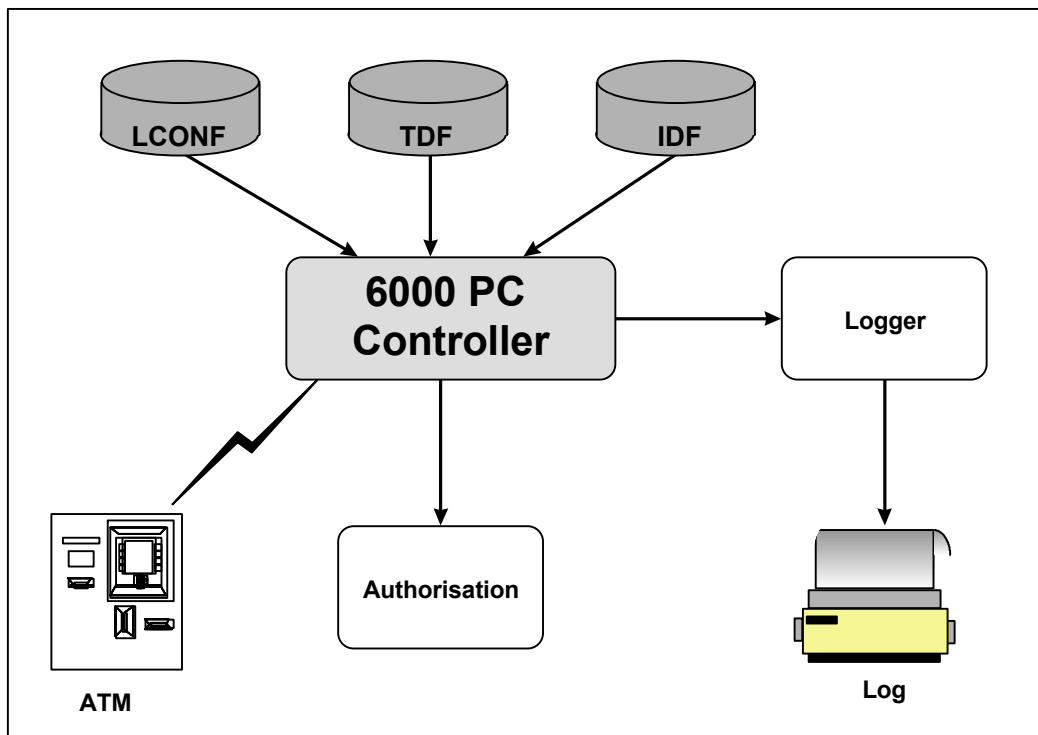


Figure 1-3 Olivetti 6000 Device Handler

1.2.1 Features Supported

The Olivetti 6000 Device Handler supports Olivetti 6000 ATM firmware Versions A, B and C. However, this guide only documents devices which support Version C. The features of the Device Handler include:

- ANSI X9.8 PAN PIN block support
- Card capture support
- Communication support for SDLC, Bisynchronous, X.25, SNAX/HLS and TCP/IP protocols
- Dynamic key changes
- Enhanced status management
- File transfer service with data compression
- Fraud processing
- Hardware MACing
- Hardware security module support
- Inactivity timers
- Message encryption in software
- Pin change transaction
- PIN/PAN PIN block support

- Printer support
 - ⇒ for journals
 - ⇒ for receipts
 - ⇒ for statements
- Setting of the date and time
- Envelope deposits
- Commercial deposits
- ATM withdrawal surcharging
- Electronic Journal
- Currency Exchange

In addition, there is support for receipt printing in release 5.1, but the Olivetti 6000 ATM already manages receipt printing. For more details see section 7 of this user guide.

1.2.2 Transaction Support

The Olivetti 6000 Device Handler supports the following customer and administrative transactions for the Olivetti 6000 series ATMs:

- Customer Transactions
- Administrative Transactions
- Status Messages.

The following sections summarise the type of transactions supported. For more details, refer to the section 5.

1.2.2.1 Customer Transactions

The customer transactions supported by the Olivetti 6000 Device Handler are:

- Deposit (using envelopes or commercial depository)
- Electronic Payment (only supported if multi-account cards (card review) is configured)
- Inquiry
- LOGONLY
- Message Enclosed
- Payment Enclosed
- PIN Change
- Statement Print
- Transfer - only supported if multi-account cards (card review) is configured
- Withdrawal
- Currency Exchange.

1.2.2.2 Administrative Transactions

The administrative transactions supported by the Olivetti 6000 Device Handler are:

- Cartridge Maintenance
- Cutoff Request (Close Period).

1.2.2.3 Status Messages

The status messages supported by the Olivetti 6000 Device Handler are:

- Terminal Totals
- Terminal Status
- Peripheral Status
- Cassette Status
- Currency Exchange Totals.

1.2.3 Receipt Support

The Olivetti 6000 ATM supports the printing of receipts. Receipt format and printing are controlled by configuration files held locally at the ATM. These configuration files support the following features:

- Configurable Receipts
- Optional Receipts
- PAN transactions
- Receipts for denied transactions.

Some data to be printed on receipts is provided by Olivetti 6000 Device Handler. Refer to Section 7 of this user guide for more details.

2. CONFIGURATION

The Olivetti 6000 ATMs are configured from the 6000 controller. In addition, a number of BASE24 files are used to configure the ATMs. These files are identified in Figure 2-1.

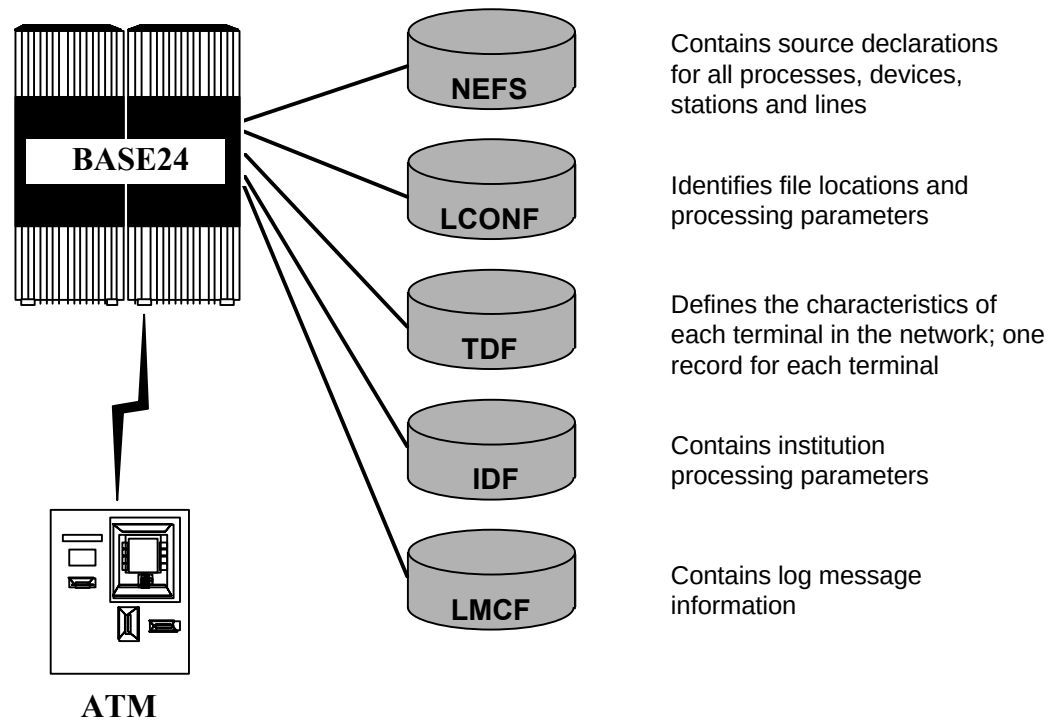


Figure 2-1 BASE24 Files

This section describes the configuration options for the:

- Network Environment Source File (NEFS)
- Logical Network Configuration File (LCONF).

The maintenance of the remaining files shown in Figure 2-1 are detailed in the section 3:

- Institution Data File (IDF)
- Log Message Configuration File (LMCF)
- Terminal Data File (TDF).

For more information on how to maintain and amend the 6000 controller configuration file, refer to the OLIVETTI 6000/BASE24 Connection Getting Started/Personalisation Guide (Release 4.0).

2.1 NETWORK ENVIRONMENT SOURCE FILE (NEFS)

The Network Environment Source File (NEFS) contains the source declaration for all processes, device types, lines and stations in each resource node. These declarations define the network components to the system. Examples are provided below of each type of NEFS declaration; that is, process, device, station and line.

2.1.1 X.25 Support

In addition to support for SNA/SDLC and bisynchronous protocols, the Olivetti 6000 Device Handler supports X.25 communications protocols for ATM communication. Both Permanent Virtual Circuits (PVCs) and Switched Virtual Circuits (SVCs) are supported.

Once the connection for PVC is established, the ATM can initiate transactions at any time and the Device Handler can send unsolicited messages at any time.

For X.25 support, the following points should be noted:

- Use of X.25 must be set via the TDF TERMINAL LINE PROTOCOL field (see section 3.6.2).
- X.25 XPNET modules are required if X.25 is used. These are the basic X.25 Network Interface module and the HOST end-to-end module.
- X.25 lines and stations must be configured in the NEFS (see sections 2.1.6 and 2.1.7).
- The LCONF parameter ATM-DH-X25-TYPE specifies the X.25 implementation—PVC, permanent SVC or ‘make and break’ SVC. This parameter is applicable to all X.25 lines/stations (and not on an individual terminal basis), refer to section 2.2.

For SVC support, the following points should be noted:

- The X.25 SVC support is a ‘point-to-point’ implementation. Each ATM must be configured with a unique X.25 Network User Address (NUA) and a sub-address. This address should correspond to the particular BASE24 line/station associated with the ATM.
- The LCONF parameter ATM-DH-DISCONNECT-X25-REQD is applicable to all X.25 lines/stations (and not on an individual terminal basis), refer to section 2.2.
- The Device Handler receives 9518 (Connect) and 9519 (Disconnect) messages from XPNET for every X.25 connection/disconnection. The logging of these messages by the Device Handler can be controlled by the LCONF parameter ATM-DH-CONNECT-MSGS. The Device Handler must be warmbooted to pick up any changes to this LCONF parameter, detailed in section 2.2.

2.1.2 SNAX/HLS Support

In addition to supporting SNA/SDLC and bisynchronous protocols, the Olivetti 6000 Device Handler supports SNAX/HLS communications protocols for ATM communication. HLS is a high level interface to Tandem's SNAX product. SNAX is an access method that implements the necessary support for communication in an SNA environment.

For SNAX/HLS support, the following points should be noted:

- Use of SNAX/HLS must be set via the TDF TERMINAL LINE PROTOCOL field (see section 3.6.2).
- SNAX/HLS lines and stations are configured in the NEFS (see sections 2.1.6 and 2.1.7).
- The BASE24 SNAX/HLS XPNET module and the Tandem SNAX/HLS product are required.

2.1.3 TCP/IP Support

Customers who require TCP/IP support should contact their local Olivetti offices.

2.1.4 Process Declarations

Below is an example of a process declaration.

```
PROCESS    P1A^OLI61:
           OBJECT                '$DATA.TES10BJ.OLI6',
           LIBRARY                '$SYSTEM.SPANNET.SKELB',
           PPD NAME IS            '$T1AM',
           FIRST DEMAND,
           PRIORITY                146,
           MAX CORE QUEUE        200,
           LOGICAL NETWORK IS    TES1,
           CPU                    3;
```

2.1.5 Device Declarations

Below are examples of device declarations.

```
DEVICE    HLS^DV:                                !HLS
           RECEIVE                                750,
           TRANSMIT                               750,
           PROFILE                                'PCOLIA',
           TRANSMISSION NUMBER LENGTH          0,
           ADDRESS CHARACTER LENGTH            2,
           UNIT ADDRESS POSITION                 2,
           XPNET RESIDENT SSCP NO;

           DEVICE                                OLI6^DV:
           RECEIVE                                750,
           TRANSMIT                               750,
```

2.1.6 Station Declarations

Below are examples of station declarations.

```

STATION    S1AHLs:                                !HLS1
           DEVICE                                HLS^DEV,
           ADDRESS                               'A': '/' ,
           LOGICAL NETWORK IS                   TES1,
           DESTINATION PROCESS                  P1A^OLI61,
           OUTPUT INTERVAL                      0,
           MAX CORE QUEUE                      2,
           TERMINAL TYPE                       0,
           RETRY                               3,
           REINSTATE AFTER                     10 SECONDS,
           SENSE STATUS QUEUE COUNT            0,
           NO LOGICAL ACK;

STATION    S1AOLI61:
           DEVICE                                OLI6^DEV,
           ADDRESS                               'D': 'U' ,
           LOGICAL NETWORK IS                   TES1,
           DESTINATION PROCESS                  P1A^OLI61,
           MAX CORE QUEUE                      50,
           TERMINAL TYPE                       0,
           RETRY                               3,
           NO LOGICAL ACK;

STATION    S1AOLI610:                             !X.25
           DEVICE                                OLI6^DEV,
           LOGICAL NETWORK IS                   TES1,
           DESTINATION PROCESS                  P1A^OLI61,
           MAX CORE QUEUE                      5,
           RETRY                               3,
           NO LOGICAL ACK,
           OUTPUT INTERVAL                     0 SECONDS,
           IGNORE ERRORS;

STATION    S1AOLI611:                             !X.25
           DEVICE                                OLI6^DEV,
           LOGICAL NETWORK IS                   TES1,
           DESTINATION PROCESS                  P1A^OLI6
           RETRY                               3,
           NO LOGICAL ACK,
           OUTPUT INTERVAL                     0 SECONDS,
           IGNORE ERRORS;

```

Note

- 1 This example is only valid for direct connection from device to Tandem.

2.1.7 Line Declarations

Below are examples of line declarations.

```

LINE      L1A^HLS                                !HLS1
          DEVICE ID                               '$SNAX1',
          PLU,
          SYNC,
          TIMEOUT                                250 TICKS,
          TIMER INTERVAL                        100 SECONDS,
          RETRY                                1,
          CONTROL CHARACTER OCCURS             2,
          UNIT CHARACTER OCCURS                2,
          POLL INTERVAL                        1 SECOND,
          OUTPUT BIAS                          3,
          RVI LEVEL                            0,
          ERROR ANALYSIS INTERVAL             1 MINUTE,
          STATIONS                             (S1AHLS);

          LINE      L1A^BISYNC                      !BISYNC
          DEVICE ID                               '$BISYNC',
          BI-SYNC MULTI-POINT SUPERVISOR,
          STATIONS                             (S1AOLI61);

LINE      L1A^OLI610                             !X.25 PVC
          DEVICE ID                               '$X25A.#DEV1',
          X25,
          SYNC,
          TIMEOUT                                5 SECONDS,
          RETRY                                0,
          PVC,
          END TO END PROTOCOL                   9,
          FULL DUPLEX
          STATIONS (                             S1AOLI610);

          LINE      L1A^OLI611                      !X.25 SVC (Make & Break)
          DEVICE ID                               '$X25A.#DEV10',
          X25,
          SYNC,
          TIMEOUT                                5 SECONDS,
          RETRY                                0,
          MAKE CALL,
          WAIT FOR CALL,
          REMOTE                                '<NUA>',
          END TO END PROTOCOL                   9,
          FULL DUPLEX
          ATIMER                                1,
          STATIONS                             (S1AOLI611);

```

Note

- 1 This example is only valid for direct connection from device to Tandem.

2.2 LOGICAL NETWORK CONFIGURATION FILE (LCONF)

The Logical Network Configuration File (LCONF) is the central location of file information and processing information for BASE24. Therefore it is one of the main components of every BASE24 Logical Network. One LCONF exists for each Logical Network.

The LCONF contains assigns which link the physical names of objects in the Logical Network to names used by BASE24 code, and parameters (params) which define processing values at the Logical Network level.

The LCONF assigns and params listed in this section are required by the BASE24 Olivetti 6000 Device Handler to initialise and process correctly.

For more details, refer to the BASE24 Logical Network Configuration File Manual.

2.2.1 General LCONF Assigns

The following assigns are used by the BASE24 Olivetti 6000 Device Handler process.

ATM-DH-OLI6-LOAD-FILE

The fully qualified Tandem file name of the upline/downline file used for file transfer; for example, \B24.\$DATA.PRO1DATA.OLI6.

This assign is used by the Device Handler processes for file transfer. If a file is not specified in the LOAD FILENAME field in the Terminal Data File (TDF) screen 8 (Figure 3-6), the upline/downline file specified in this assign is the default file.

Process Name Required if the 6000 Device Handler processes do not use the same upline/downline file.

IDF

The fully qualified file name of the Institution Definition File (IDF); for example, \B24.\$DATA.PRO1DATA.IDF. The IDF assign is used by the following processes:

- ⇒ DCT server
- ⇒ IDF server
- ⇒ BASE24-atm Authorisation process
- ⇒ BASE24-atm Device Handler process
- ⇒ BASE24-atm report programs
- ⇒ BASE24-atm Settlement Initiator process

If this assign is not present in the LCONF or contains an invalid value, the processes listed above abend at initialisation.

Product Use ATM, BASE

LMCF

The fully qualified file name of the Log Message Configuration File (LMCF); for example, \B24.\$DATA.PRODCNTL.LMCF. The LMCF assign is used by all application processes and, optionally, any XPNET processes that alter log messages.

Product Use ATM, BASE

PLF

The fully qualified file name of the PCS Log File (PLF).

SECURE-DEV1

The name of the primary security module for PIN management, message authentication processing and security module key management. For security devices other than the Atalla A6000, this assign specifies the Tandem SYSGEN device name (for example, \B24.\$SECDVA). For Atalla A6000 security devices, this assign specifies the Process Pair Directory (PPD) name for the A6000 interface module used with the security device (for example, \B24.\$NSP1). The device name or interface module name must be compatible with the Tandem Expand product.

The SECURE-DEV1 assign is used by security procedures bound into the following processes:

- ⇒ BASE24-atm Authorisation process
- ⇒ BASE24-atm Device Handler process

Product Use	ATM, BASE
Process Name	Required if each process uses a different physical device as its primary security module (for example, P1A^AUTH01, P2B^BASE02 or P2A^SPDH03).
User Field	Required for Atalla Security Modules (ASMs) and Atalla Network Security Processors (NSPs). For ASMs, the value in the USER FIELD fields indicates the model type (for example, A1, A2 or A3). For both NSP models, A5000 and A6000, the value in the USER FIELD is NSP.

SECURE-DEV2

The name of the secondary security module for PIN management, message authentication processing and security module key management. For security devices other than the Atalla A6000, this assign specifies the Tandem SYSGEN device name (for example, \B24.\$SECDVB). For Atalla A6000 security devices, this assign specifies the Process Pair Directory (PPD) name for the A6000 interface module used with the security device (for example, \B24.\$NSP2). The device name or interface module name must be compatible with the Tandem Expand product.

The SECURE-DEV2 assign is used by security procedures bound into the following processes:

- ⇒ BASE24-atm Authorisation process
- ⇒ BASE24-atm Device Handler process

Product Use	ATM, BASE
Process Name	Required if each process uses a different physical device as its secondary security module (for example, P1A^AUTH01, P2B^BASE02 or P2A^SPDH03).
User Field	Required for Atalla Security Modules (ASMs) and Atalla Network Security Processors (NSPs). For ASMs, the value in the USER FIELD indicates the model type (for example, A1, A2 or A3). For both NSP models, A5000 and A6000, the value in the USER FIELD is NSP.

TDF

The fully qualified file name of the Terminal Data File (TDF); for example, \B24.\$DATA.PRO1DATA.TDF. The TDF assign is used by the following processes:

- ⇒ TDF server
- ⇒ BASE24-atm Device Handler process
- ⇒ BASE24-atm Settlement Initiator process.

2.2.2 General LCONF Parameters

The following params are used by the BASE24 Olivetti 6000 Device Handler process.

ATM-DH-AUTH-TIMER

The number of seconds the Device Handler process waits, after sending a 0200 request message, for a reply from the BASE24-atm Authorisation process before denying a request. This value must be less than the device timeout so that the Device Handler can send a timeout response to the device prior to the device timing out. The valid values are 10-180 seconds.

The BASE24-atm Device Handler processes use this param.

Default 45 (seconds)

ATM-DH-CONNECT-MSG

The protocol for ATM communication is X.25. The valid values are:

ON The 6000 Device Handler logs the ATM Up and ATM Down messages.
OFF The ATM Up and ATM Down messages are not logged.

Default ON

ATM-DH-CONS-MAC-ERR-LMT

The number of consecutive security MAC errors that can occur before the Device Handler process loads a new MAC key. The valid values are 0 - 32767. The value 0 indicates that the Device Handler process does not shut down the ATM regardless of the number of consecutive security MAC errors. These errors are not accumulated during file transfer.

Process Name Required if the limit varies among Device Handler processes
Default 3

ATM-DH-DISCONNECT-X25-REQD

The protocol for ATM communication is X.25 and this parameter is applied globally to all X.25 connected ATMs. The valid values are:

YES The Device Handler instructs XPNET to disconnect an X.25 circuit, if required. Disconnection is required after a message has been received from the ATM that does not require BASE24 to respond.
NO BASE24 will not disconnect X.25 circuits. The ATM is required to disconnect the X.25 circuit, when required.

This parameter should be set to 'NO' in the case of PVC or SVC. The XPNET ATIMER can be used instead to disconnect the circuit after an idle period.

Default NO

ATM-DH-MAX-IN-CORE

The maximum number of Terminal Data File (TDF) records that a Device Handler process can store in memory at one time. This param represents the amount of extended memory each Device Handler process must reserve for TDF records and the maximum number of transactions each Device Handler process can process at one time.

The Device Handler process stores a TDF record in memory each time it receives a transaction from an ATM. The Device Handler process retains that TDF record in memory until it receives an authorisation response from the Authorisation process.

If the value in this param is too low, the Device Handler process may run out of extended memory during peak processing times and transactions are declined until more extended memory becomes available. If the value in this param is set too high, the excess extended memory is wasted.

The value in this param should be an estimate of the maximum number of devices that can have transactions in progress at one time. However, the value in this param should not be greater than the value in the ATM-DH-MAX-TDFS param. The valid values are 1-32767 records.

The BASE24-atm Device Handler processes use this parameter.

Process Name	Required if this number varies among BASE24-atm Device Handler processes.
Default	25 (records)

ATM-DH-MAX-TDFS

The maximum number of Terminal Data File (TDF) record keys that each BASE24-atm Device Handler process can store in memory. This param represents the maximum number of terminals each Device Handler process can service.

When a Device Handler process initialises or reinitialises, space is allocated in memory to build a table. This table stores the record positions of all TDF records that the Device Handler process can access. By accessing this table, the Device Handler process can quickly locate the devices to which it must send transactions.

If the value in this param is too low, the Device Handler process may run out of space in the table and be unable to process for all the devices. If the value in this param is set too high, the excess table space is wasted.

The value in this param should be equal to the total number of devices that are associated with each Device Handler process. However, the value in this param should be greater than the value in the ATM-DH-MAX-IN-CORE param. The valid values are 1-32767 records.

The BASE24-atm Device Handler processes use this parameter.

Process Name	Required if the number of ATMs associated with a BASE24-atm Device Handler process varies among BASE24-atm Device Handler processes.
Default	100 (records)

ATM-DH-OLI-CE-OLI-CE-ACTIVITY

Used by the Device Handler to enable logging of the A06 message in full. The valid values are:

- 0 No logging of 'A06' messages (default)
- 1 Log 'A06' messages

If the parameter is not present or invalid the Device Handler will not log A06 messages.

ATM-DH-OLI6-CMD-TIMEOUT

The number of seconds the 6000 Device Handler process allows a file transfer request, security data maintenance request or DCT command to complete. For file transfer requests and security data maintenance requests, this timer is set only if a load is in progress. The valid values are 10 to 180 seconds.

Default 45 (seconds)

ATM-DH-OLI6-KEY-CHG-TIME

The time after which the PIN Block and MAC keys are automatically changed daily. The format is HHMM.

A dynamic key change is performed daily following the first approved withdrawal transaction after the specified time.

If this param is absent then no dynamic key changes will be performed. To change the dynamic key change time after the next dynamic key change has taken place, perform a warmboot. The Device Handler reads the param, but the timer will not be reinitialised. The new time will be effective when the current timer has expired.

Default No default value

ATM-DH-OLI6-LOAD-MSG-LEN

The maximum Data Block Size (DBS) to be transferred in each G01/g01 message. The valid values are 256-3920 bytes and must be divisible by 16.

Default 256

ATM-DH-OLI6-POLL-INTERVAL

The number of seconds the Device Handler allows before initiating a terminal status poll request if no messages have been received from the ATM.

Default 300 (seconds)

ATM-DH-OLI6-POLL-RETRY-MAX

The maximum number of times the Device Handler retries sending a terminal status request to the ATM without receiving a response, before marking the ATM as HUNG. A response is deemed to be either a terminal status response message or any other message.

Default 3 (retries)

ATM-DH-OLI6-RETRY-INTERVAL

The number of seconds the Device Handler allows between timing out a response to a terminal status request and retrying. A response is deemed to be either a terminal status response message or any other message. Retry processing is performed for status requests initiated by the Device Handler as poll messages and for status requests initiated from DCT screen 32 as STATUS messages.

Default 60 (seconds)

ATM-DH-OLI6-TERM-STATUS

The logging of the contents of the A05 and A0R messages in full by the Device Handler. If this parameter is not present or invalid, the Device Handler will not log A05 and A0R messages. The valid values are:

- 0 No logging of A05 and A0R messages
- 1 Log A05 messages only
- 2 Log A0R messages only
- 3 Log A05 and A0R messages

Default 0 (No logging of A05 and A0R messages)

ATM-DH-STALE-RQST-TIMEOUT

The number of seconds an incoming transaction request remains in the BASE24-atm Device Handler process queue before it is discarded. The valid values are 10-180 seconds. The recommended setting is 20 (seconds).

The BASE24-atm Device Handler processes use this parameter.

Process Name Required if the number of seconds an incoming transaction request can remain in the Device Handler queue varies among BASE24-atm Device Handler processes.

Default 45 (seconds)

ATM-DH-X25-TYPE

Specifies the type of X.25 protocol being used. The valid values are:

- | | |
|---|-------------------------|
| 0 | X.25 PVC |
| 1 | X.25 Permanent SVC |
| 2 | X.25 Make and Break SVC |

Default 0 (X.25 PVC)

SECURE-DEV-FAIL-NOTIFY

The interval at which critical messages are logged due to the failure of a process primary or secondary security modules. The valid values are 10-1000.

If a process primary and secondary security module cannot be initialised during the process startup and this param is set to the default of 20, every 20th transaction that a process must decline because the security modules are inoperative causes a message to be logged indicating the security modules have failed. A critical log message is always generated immediately upon failure of both devices.

If a process primary security module is marked down and this param is set to the default of 20, every 20th transaction that is directed to the primary security module causes a message to be logged even though the transactions are handled properly by the secondary security module. Transactions routed directly to the secondary security module do not affect the primary security module transaction counter. The SECURE-DEV-MONITOR-THRESHOLD param controls when a transaction is directed to the secondary security module instead of the primary security module.

If a process primary and secondary security module fails at some time after initialisation, a separate failed transaction counter is maintained for each security module. A transaction that is sent to one security module and then the other increments both counters. A message is logged each time either counter reaches the value established in this parameter.

The SECURE-DEV-FAIL-NOTIFY param is used by security procedures bound into the following processes:

- ⇒ BASE24-atm Device Handler
- ⇒ BASE24-atm Authorisation process

Product Use ATM, BASE
Default 20

SECURE-DEV-TIM-LMT

The maximum time, in 100ths of a second BASE24 waits for a response from a security device before the message times out. The valid values are 1-6000 hundredths of a second.

The SECURE-DEV-TIM-LMT parameter is used by security procedures bound into the following processes:

- ⇒ BASE24-atm Device Handler
- ⇒ BASE24-atm Authorisation process

Product Use	ATM, BASE
Default	500 (5 seconds)

SECURE-DEV-TYP

A code identifying the type of security module specified in the SECURE-DEV1 and SECURE-DEV2 assigns. The valid values are:

ASM	Atalla Security Module or Atalla Network Security Processor
RSM	Racal-Guardata Host Security Module (the same as VSM)
VSM	Racal-Guardata Host Security Module (the same as RSM)
N/A	A security module is not required

The SECURE-DEV-TYP parameter is used by security procedures bound into the following processes:

- ⇒ BASE24-atm Device Handler
- ⇒ BASE24-atm Authorisation process

Product Use	ATM, BASE
Default	No default value.

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3. FILE ACCESS & MAINTENANCE

The Olivetti 6000 device is configured via the ATM configuration file held in the ATM. This file contains the control logic and associated parameters that determine how transaction processing is configured at the ATM level; for example, screen data.

For more information on how to maintain and amend the 6000 controller configuration file, refer to the OLIVETTI/BASE24 Connection Getting Started/Personalisation Guide (Release 4.0).

In addition to the ATM configuration file, a number of BASE24 files are used to configure transaction processing at the network level to support the 6000 ATMs. BASE24 files provide indirect support to the ATM by defining the interaction between the Device Handler and the ATM.

Because of the many combinations of options that are available during configuration, the BASE24 files are discussed in this section. The screen descriptions detail the fields on the screen, the valid values that can be placed in the fields during configuration, as well as the default values and lengths.

Function keys that operate differently from the standard are also documented.

For more details, refer to the BASE24 Files Maintenance Manual.

3.1 SCREEN LAYOUT & KEYBOARD CONVENTIONS

3.1.1 Standard Screen Layout

File maintenance screens display the following header line at the top of each screen display:

```
<prod>  <type>                LLLL  FIID  YY/MM/DD  HH:MM  PAGE nn OF nn
```

where:

'<prod>'	specifies the screens as part of the BASE24 system for the product specified.
'<type>'	specifies the type of screen displayed.
'LLLL'	specifies the Logical Network and displays the Logical Network Identifier entered at individual operator Logon.
'FIID'	specifies the identification of the financial institution.
'YY/MM/DD'	specifies the current system processing date. This field is displayed whenever an AFT screen is accessed. The format of the field is configurable using the LCONF parameter, DATE-DISPLAY-FORMAT.
'HH:MM'	specifies the current system processing time at the time the screen was displayed. This field is displayed whenever an AFT screen is accessed.
'nn OF nn'	specifies the screen number of the screen currently displayed as well as the total number of screens for that particular file.

3.1.2 Standard Function Keys

The standard BASE24 function keys are used on BASE24 screens. When the usage of these function keys varies with BASE24 files and functions, the differences are noted with the screen descriptions for the file.

BASE24 has an Online Help Screen which lists the function keys that apply on each screen. The Help Screen is displayed by pressing the F12 function key.

The usage of the function keys displayed on the screen is explained in the following table. The first column shows the function key. The second and third columns provide a description of how the function key works for the screen.

KEYS	FUNCTION	DESCRIPTION
F1	VALIDATE DATA	Checks that the entered data conforms to the specified standards or values.
F2	READ RECORD	Reads a record from the current file.
F3	ADD RECORD	Adds a unique record to the current file.
F4	DELETE RECORD	Deletes a record from the current file.
F5	UPDATE RECORD	Changes an existing record in the current file.
F6	READ NEXT RECORD	Reads the next record in the current file.
F7	GOTO NEW PAGE	Displays a different screen in the current file.
F8	CLEAR SCREEN	Clears any values in the record to spaces or replaces them with default values. This impacts all screens associated with the file accessed including the current screen, any previous screens, and any remaining screens. Once this function has been used, the update function cannot be completed until the read/read next function is completed.
F9	NEXT SCREEN	Displays the next screen of the file accessed. If the last screen was displayed and this function key was pressed, the first screen of the file will be displayed.
F10	PRINT SCREEN	Prints a hard copy of the displayed screen. If a printer is not specified and the system default is used, the print job will remain in the spooler until the location/status is changed by the operator.
F11	PREVIOUS SCREEN	Displays the previous screen of the file accessed. If the first screen was displayed and this function key was pressed, the last screen of the file will be displayed.
F12	DISPLAY HELP SCREEN	Displays function key information or menu options.
F13	CHANGE CURRENT LOGICAL NET	Changes the current logical network while in a file.
F16	EXIT OR GOTO FILE	This key has several purposes. It is used to exit BASE24 or the file currently accessed. It also enables the user to move between logical networks and files.
SF16	LOGOFF	Ends a session as a terminal or operator.

3.2 USING ONLINE HELP

3.2.1 Example of a Help Screen

All information displayed on the Help Screen Figure 3-1 is read only.

To exit from the Help Screen, press either any key (with the exception of SF9 - SF16) or F16 to display the screen from which the F12 (HELP) function key was pressed.

Figure 3-1 an example of a standard Help Screen.

```

BASE24-ATM  TERMINAL DATA  REL1  96/06/28 15:34 01 OF 12
TERM ID:      FIID:      REGION:  BRANCH:  LN: REL1
STANDARD FILE MAINTENANCE FUNCTION FOR KEYS 6520/6530 (3270)
F1 (ENTER) - VALIDATE DATA
F2 (PF2)   - READ RECORD
F3 (PF3)   - ADD RECORD
F4 (PF4)   - DELETE RECORD
F5 (PF5)   - UPDATE RECORD
F6 (PF6)   - READ NEXT RECORD
F7 (PF7)   - GOTO NEW PAGE
F8 (PF8)   - CLEAR SCREEN
F9 (PF9)   - NEXT SCREEN
F10 (PF10) - PRINT SCREEN
F11 (PF11) - PREVIOUS SCREEN
F12 (PF12) - DISPLAY HELP SCREEN
F13 (PF13) - CHANGE CURRENT LOGICAL NET
F16 (PF16) - EXIT OR GOTO FILE
SF16 (PA2) - LOGOFF

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:  NEW LOGICAL NETWORK ID:
ANY FUNCTION KEY EXCEPT SF9-SF16, F10, OR F16 RETURNS. F10 PRINTS AND RETURNS
BLOCK
  
```

Figure 3-1 Example Help Screen

3.3 FILE ACCESS

All the BASE24 product files and functions that can be accessed are listed on the BASE24 Product Menu, accessed from the Virtual Menu.

Only the files and screen groupings available for a particular logon will be displayed for that logon.

3.3.1 Virtual Menu

The BASE24 Virtual Menu gives the operator access to all the BASE24 products the institution has purchased, assuming that the operator has the appropriate security.

Figure 3-2 is an example of a Virtual Menu screen.

BASE24-ADMM	VIRTUAL MENU	REL1	96/06/28	14:56	01 OF 01
V I R T U A L M E N U					
BASE POS DCT SWITCH FUNCTIONS			ATM SPAN CONTROL OPERATOR FUNCTIONS		
***** BASE24 *****					
FILE DESTINATION					
F1-ENTER DATA	F10-PRINT	F16-EXIT	SF16-LOGOFF		
BLOCK					

Figure 3-2 Example Virtual Menu Screen

3.3.2 BASE24 Product Menu

To display a particular BASE24 Product Menu screen, select the option by placing the cursor on the required item on the Virtual Menu and press the F1 function key. The specified BASE24 Product Menu screen will be displayed.

All BASE24-atm files accessible to the operator are listed on the ATM Product Menu, which can be accessed from the BASE24 Virtual Menu. Each operator only sees the files available which are based on the configuration of the security record.

Figure 3-3 is an example of a BASE24-atm Product Menu screen. The order in which the files are displayed for individual operators is dependent on the order in which the operator received access to the files.

BASE24-ADMIN MENU		REC1		94/08/19 19:18 01 OF 01	
A T M P R O D U C T M E N U					
BSF		CCF		CSF OIF	
RCPT		RPT		TDF TLF	
***** BASE24 *****					
FILE DESTINATION:					
F1-ENTER DATA		F10-PRINT		F12-HELP F16-EXIT SF16-LOGOFF	

Figure 3-3 Example BASE24-atm Product Menu

3.4 INSTITUTION DEFINITION FILE (IDF)

The Institution Definition File (IDF) contains one record for each institution participating in the logical network and defines processing for each institution.

The IDF contains routing tables for transaction routing in BASE24 and each institution's parameters for cards, dates, ATM processing control and sharing.

The key to records in the IDF is the FIID for the financial institution.

The IDF is read by the Device Handler process during an administrative transaction to retrieve the dates the Device Handler process requires for cutting over terminals (IDF screen 8).

3.5 LOG MESSAGE CONFIGURATION FILE (LMCF)

The Log Message Configuration File (LMCF) contains one record for each log message. LMCF records must be read into memory by the processes using them, thus the updates to the LMCF do not take effect until the processes are reinitialised.

An LMCF record is accessed via the BASE24-atm operator functions. The file contains changes made to the log messages, in the form of records. Each log message routing code, severity code and text can be modified via the BASE24 operator functions.

For further information on BASE24 operator functions and how to access the LMCF, refer to the BASE24 Network Control Operators Manual.

3.6 TERMINAL DATA FILE (TDF)

The Terminal Data File (TDF) contains one record for each ATM terminal in the logical network. An ATM terminal is also referred to as an ATM device. Whenever BASE24 receives a message from an ATM, or a message is sent to an ATM, the TDF record is accessed.

The TDF defines the characteristics of the terminals, including the transaction security parameters, sharing parameters, transaction totals, terminal owner and device-dependent data. It is also used to preserve transaction context and is updated every time a transaction is processed for the terminal.

The first six TDF screens contain information which is relevant regardless of the device type used. The TDF screens 7, 8 and 9 contain device-specific information and these screens are not used with all device types.

This section describes the TDF screens that have been changed to provide specific support for the Olivetti 6000 series ATMs. These screens are:

- **TDF screens 4 and 5**—contain information on the hopper contents (the cash value in each hopper in the terminal).
- **TDF Olivetti 6000 screen 7**—contains information specific to 6000 terminals and inactivity timer information (to monitor transaction activity at each ATM).
- **TDF Olivetti 6000 screen 8**—contains information on the control of receipts at the 6000 ATM.
- **TDF Olivetti 6000 screen 9**—contains information on 6000 security, PIN details and message authentication.

Note!

On the TDF screen 6 (not documented in this guide), the amount of the debit fields reflect the TERMINAL CREDITS field amount which is a total of withdrawals, transfers and payments from any account.

For further information on maintaining TDF screens, refer to the BASE24-atm Files Maintenance Manual.

3.6.1 TDF Screens 4 and 5

The Terminal Data File (TDF) screens 4 and 5 contain hopper totals. Since the screens are identical except for the hopper numbers, only screen 4 is shown (Figure 3-4).

The TDF screen 4 contains information for hoppers 1, 2 and 3, and the TDF screen 5 contains information for hoppers 4, 5 and 6.

For Olivetti 6000 series ATMs these screens remain unchanged, but they are used in a slightly different way to that described in the BASE24-atm Files Maintenance manual. Only hopper 1 is used, and reflects the total value of all hoppers in the ATM; hoppers 2, 3, 4, 5 and 6 are still shown, but are not used. In addition, the field BILL VAL, for hopper 1, must be set to the value 1. In all other respects the screens remain unchanged.

The field descriptions that follow should be read in conjunction with the notes on terminal balancing. Refer to the section 5.2 for more details on terminal balancing.

BASE24-ATM TERMINAL DATA		REL1 OLIV 96/07/05 16:15 04 OF 12	
TERM ID: S1A0LI6		FIID: OLIV REGION: 0001 BRANCH: 0001 LN: REL1	
HOPPER CONTENTS			
	HOPPER 1	HOPPER 2	HOPPER 3
	00 (CASH)	00 (CASH)	00 (CASH)
CONTENTS:			
BILL VAL:			
STD BEG CASH:			
BEG CASH:			
CASH INCR:			
CASH DECR:			
CASH OUT:			
END CASH:			
CURRENCY CODE:	826 (UK)	826 (UK)	826 (UK)
***** BASE24 *****			
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:	
	F12-HELP		
BLOCK			

Figure 3-4 TDF Screen 4

TERM ID**16 (AN) M**

An identifier for the terminal associated with this record. Each entry in the TERM ID field must be unique in the logical network. An attempt to add a duplicate TERM ID will be rejected. The TERM ID field is an alternate key to the TDF.

The identifier entered must match a symbolic station name defined in the Network Environment Source File (NEFS). Station names are defined in the STATION IDENTIFICATION element of the NEFS (see section 2.1.6).

Default Value	No default value
Data Name	TDF.TERM-ID

FIID**4 (AN) M**

The FIID of the financial institution that owns the terminal. The FIID entered in this field must be defined in the Institution Definition File (IDF).

The REGION, BRANCH and LN fields combine with the FIID field to make one alternate key to the TDF.

Default Value	The FIID previously entered
Data Name	TDF.TERM-OWNER.FIID

REGION**4 (AN)**

The region identifier for the region in which this terminal is located. This field is used in conjunction with the Device Control Terminal (DCT) functions (detailed in section 4). Although an entry is not required to add a record, it is recommended that an entry is made.

The BRANCH, LN and FIID fields combine with the REGION field to make one alternate key to the TDF.

Default Value	No default value
Data Name	TDF.TERM-OWNER.REGN-ID

BRANCH**4 (AN)**

The branch identifier for the branch in which this terminal is located. This field is used in conjunction with the Device Control Terminal (DCT) functions (detailed in section 4). Although an entry is not required to add a record, it is recommended that an entry is made.

The LN, FIID and REGION fields combine with the BRANCH field to make one alternate key to the TDF.

Default Value	No default value
Data Name	TDF.TERM-OWNER.BRCH-ID

LN**Display Only**

The name of the logical network to which the terminal owner belongs.

The FIID, REGION and BRANCH fields combine with the LN field to make one alternate key to the TDF.

When adding a new record, BASE24-atm sets the value in this field to the logical network of the person performing the file maintenance.

Data Name TDF.TERM-OWNER.LN

HOPPER CONTENTS

The following fields define the cash value in each hopper in the terminal. HOPPER 1 always contains the lowest bill denomination, with each successive hopper containing higher bill denominations. For Olivetti 6000 series ATMs, the ATM itself manages the hoppers. HOPPER 1 is the only hopper which is configured.

CONTENTS**2 (N) M**

A code specifying what is contained in each hopper. The valid codes are:

00	Cash (Default)
01	Coins

Additional values can be defined by the terminal owner.

Data Name TDF.HOPR.CONTENTS

BILL VAL**7 (N)**

A value identifying the denomination of bills in the hopper.

For Olivetti 6000 series ATMs, the ATM itself manages the hoppers. Hence, the value of this field must be 1 for Hopper 1. No other hoppers can be configured.

Default Value	No default value
Data Name	TDF.HOPR.BILL-VAL

STD BEG CASH**15 (N)**

The standard amount of money used to replenish cartridge cash dispensers. The standard amount of money must be divisible by the value entered in the BILL VAL field.

The STD BEG CASH field is only used with terminals which utilise cartridge cash dispensers. With these types of dispensers, a new cartridge, which contains the standard amount of money, is inserted in the hopper to replace the previous cartridge. This field is used when balancing a terminal.

Default Value	No default value
Data Name	TDF.HOPR.STD-BEG-CASH

BEG CASH**15 (N)**

The amount of money in the hopper at the start of the current balancing period. The amount of money must be divisible by the value entered in the BILL VAL field.

The BEG CASH field is used when balancing the terminal to current cash. It can differ each time the terminal is balanced. The first amount to be entered in this field is entered by the operator via this screen. Thereafter, BASE24 automatically maintains these amounts.

Default Value	No default value
Data Name	TDF.HOPR.BEG-CASH

CASH INCR**15 (N)**

The amount of money added to the hopper through administrative mid-point adjustment transactions since the last balancing transaction. The amount of money must be divisible by the value entered in the BILL VAL field.

When balancing with standard beginning cash, an amount is entered in this field initially via this screen. Thereafter, BASE24 automatically maintains this field in conjunction with administrative transactions.

When balancing with either current cash or standard beginning cash, the CASH INCR field is reset to zero.

Example	6000 = \$6000-00 (if US currency is used)
Default Value	No default value
Data Name	TDF.HOPR.CASH-INCR

CASH DECR**15 (N)**

The amount of money removed from the hopper through administrative mid-point adjustment transactions and currency exchange transactions since the last balancing transaction. The amount of money must be divisible by the value entered in the BILL VAL field.

If balancing with standard beginning cash, an amount is entered in this field initially via this screen. Thereafter, BASE24 automatically maintains this field in conjunction with administrative transactions and currency exchange status transactions.

When balancing with either current cash or standard beginning cash, the CASH DECR field is reset to zero.

Default Value	No default value
Data Name	TDF.HOPR.CASH-DECR

CASH OUT**15 (N)**

The amount of money dispensed from the hopper through customer withdrawals between terminal balancing procedures. BASE24 automatically maintains this field. When balancing with either current cash or standard beginning cash, the CASH OUT field is reset to zero.

Example	240 = \$240-00 (if US currency is used)
Default Value	No default value
Data Name	TDF.HOPR.CASH-OUT

END CASH**15 (N)**

The amount of money in this hopper. The amount in the END CASH field is entered initially via this screen. Thereafter, BASE24 automatically maintains the current cash amount in this field.

When balancing with the current cash method, the amount in the END CASH field is moved to the BEG CASH field (but is not cleared). When balancing with the standard beginning cash method, the amount in the STD BEG CASH field is moved to the BEG CASH field and the END CASH fields.

Example	45000 = \$45 000-00 (if US currency is used)
Default Value	No default value
Data Name	TDF.HOPR.END-CASH

CURRENCY CODE**3 (N) M**

The ISO Standard Code identifying the currency in the hopper. The valid values are listed in the Codes for the Representation of Currencies and Funds booklet, ISO 4217-1990.

Default Value	The default value depends upon configuration variables.
Data Name	TDF.HOPR.CRNCY-CDE

3.6.2 TDF Olivetti 6000 Screen 7

The Terminal Data File (TDF) Olivetti 6000 screen 7 is displayed for Olivetti terminals. The DEVICE TYPE field on the TDF screen 1 must be set to 32 (Olivetti Device).

The fields displayed on the TDF Olivetti 6000 screen 7 (Figure 3-5) are explained below.

BASE24-ATM	TERMINAL DATA	REL1 OLIV	96/06/28	15:56	07 OF 12
TERM ID:	FIID: OLIV	REGION:	BRANCH:	LN: REL1	
OLIVETTI INFORMATION					
TERMINAL LINE PROTOCOL: X (X.25)					
VERSION ID: C					
DEPOSITORY TYPE: 0 (ENVELOPE)					
INACTIVITY TIMER INFORMATION					
WORKDAY PEAK TIME 00:00 TO 00:00 INACTIVITY TIME-OUT 00 (MINUTES)					
WEEKEND/HOLIDAY PEAK TIME 00:00 TO 00:00 INACTIVITY TIME-OUT 00 (MINUTES)					
TIMEOUTS ALLOWED: 00 NON-PEAK INACTIVITY TIME-OUT 00 (MINUTES)					
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

Figure 3-5 TDF 6000 Screen 7

OLIVETTI 6000 INFORMATION

The following fields contain information specific to the Olivetti 6000 terminals.

TERMINAL LINE PROTOCOL**1 (A) M**

A code indicating the line protocol for communication with this terminal. The valid values are:

A	Async
B	Bisync (EBCDIC - Default)
C	Bisync (ASCII)
H	SNAX/HLS
S	SNA/SDLC
X	X.25
T	TCP/IP

Data Name TDF.MIRR.TERM-PROTO

VERSION ID**1 (A) M**

A code indicating the current version of the ATM internal software. This code is included in every message to and from the ATM. The 6000 ATM device checks the version sent by the Olivetti Device Handler. The signon sequence will fail if the value is invalid. The valid values are A, B and C.

Default Value No default value
Data Name TDF.MIRR.VERSION-ID

DEPOSITORY TYPE**1 (N) M**

The following codes indicate the Depository type configured at the ATM. The valid codes are:

0	Envelope
1	Commercial
2	Both

Default Value No default value
Data Name TDF.MIRR.AVAIL-DEP-TYP

INACTIVITY TIMER INFORMATION

The following fields are used to monitor transaction activity at each ATM. The Device Handler keeps timers for each ATM and logs messages when a transaction has not been initiated at a terminal for the specified time interval. Windows can be defined for peak and non-peak cycles as well as for holiday schedules.

The Device Handler uses the holiday table from the Institution Definition File (IDF) record (screen 4) when setting and processing the inactivity timers for holiday periods.

Existing/New inactivity timers are not deleted/added during a warmboot. The new peak periods and peak intervals are used when an inactivity timer expires or is deleted and reinserted due to the processing of a message from the Device Control Table (DCT) or the 6000 ATM device.

WORKDAY PEAK TIME (START) 4 (N)

The start time, expressed in the ATM local time, for the peak period of the work day when inter-transaction timing is to take place.

If either the WORKDAY PEAK TIME (START) or WORKDAY PEAK TIME (END) fields are set to 0000, no peak inactivity timer processing will be performed. The valid values are 0000 - 2359.

Required Field	Yes, for peak inactivity timer processing to occur
Default Value	0000
Data Name	TDF.MIRR.INTERTRANSACTION.WRKDAY-STRT-PEAK-WINDOW

WORKDAY PEAK TIME (END) 4 (N)

The end time, expressed in the ATM local time, for the peak period of the work day when inter-transaction timing is to take place.

If either the WORKDAY PEAK TIME (START) or WORKDAY PEAK TIME (END) fields are set to 0000, no peak inactivity timer processing will be performed. The valid values are 0000 - 2359.

Required Field	Yes, for peak inactivity timer processing to occur
Default Value	0000
Data Name	TDF.MIRR.INTERTRANSACTION.WRKDAY-END-PEAK-WINDOW

(WORKDAY) INACTIVITY TIME-OUT 2 (N)

The amount of time, in minutes, to use when setting the inactivity timer during the work day peak time.

For peak activity processing to occur, this field must contain a value greater than 00 when the WORKDAY PEAK TIME (START and END) fields contain values greater than 0000. The valid values are 00 - 99.

Required Field	Yes, if the WORKDAY PEAK TIME (START and END) fields contain values greater than 0000
Default Value	00
Data Name	TDF.MIRR.INTERTRANSACTION.WRKDAY-PEAK-INTERVAL

WEEKEND/HOLIDAY PEAK TIME (START)**4 (N)**

The start time, expressed in the ATM local time, of the weekend/holiday peak period in which inter-transaction timing is to take place.

If either the WEEKEND/HOLIDAY PEAK TIME (START) or the WEEKEND/HOLIDAY PEAK TIME (END) fields are set to 0000, no weekend/holiday peak inactivity timer processing will be performed. The valid values are 0000 - 2359.

Required Field	Yes, for weekend/holiday peak inactivity timer processing to occur
Default Value	0000
Data Name	TDF.MIRR.INTERTRANSACTION.WKND-STRT-PEAK-WINDOW

WEEKEND/HOLIDAY PEAK TIME (END)**4 (N)**

The end time, expressed in the ATM local time, of the weekend/holiday peak period in which inter-transaction timing is to take place.

If either the WEEKEND/HOLIDAY PEAK TIME (START) field or the WEEKEND/HOLIDAY PEAK TIME (END) fields are set to 0000, no weekend/holiday peak inactivity timer processing will be performed. The valid values are 0000 - 2359.

Required Field	Yes, for weekend/holiday peak inactivity timer processing to occur
Default Value	0000
Data Name	TDF.MIRR.INTERTRANSACTION.WKND-END-PEAK-WINDOW

(WEEKEND/HOLIDAY) INACTIVITY TIME-OUT**2 (N)**

The amount of time, in minutes, to use when setting the inactivity timer during the weekend/holiday peak time.

For weekend/holiday peak inactivity timer processing to occur, this field must contain a value greater than 00 when the WEEKEND/HOLIDAY PEAK TIME (START and END) fields contain values greater than 0000. The valid values are 00 - 99.

Required Field	Yes, when the WEEKEND/HOLIDAY PEAK TIME (START and END) fields contain values greater than 0000
Default Value	00
Data Name	TDF.MIRR.INTERTRANSACTION.WKND-PEAK-INTERVAL

TIMEOUTS ALLOWED**2 (N) M**

The number of consecutive timeouts to allow before the Device Handler process stops timing inactivity from this device. The valid values are 00 - 99.

Default Value	00
Data Name	TDF.MIRR.INTERTRANSACTION.TIMEOUTS-ALLWD

NON-PEAK INACTIVITY TIME-OUT

2 (N) M

The amount of time, in minutes, to use when setting the inactivity timer during the non-peak time (not in either of the defined peak periods).The valid values are 00 - 99.

Default Value	00
Data Name	TDF.MIRR.INTERTRANSACTION.NON-PEAK-INTERVAL

3.6.3 TDF OLIVETTI 6000 Screen 8

The Terminal Data File (TDF) Olivetti 6000 screen 8 is displayed for Olivetti 6000 terminals. The DEVICE TYPE field on TDF screen 1 must be set to 32 (Olivetti 6000 Device).

The fields displayed on the TDF Olivetti 6000 screen 8 (Figure 3-6) are explained below.

BASE24-ATM	TERMINAL DATA	REL1 OLIV	96/06/28	16:02	08 OF 12
TERM ID:	FIID: OLIV	REGION:	BRANCH:	LN: REL1	
OLIVETTI RECEIPT INFORMATION					
CUSTOMER BALANCE INFORMATION: 0 (USE BOTH AMTS)					
CUSTOMER BALANCE DISPLAY: 0 (DISPLAY ON SCREEN)					
AUDIT PRINTER OPTION: 2 (NO AUDIT PRINTER)					
LOAD FILENAME:					
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

Figure 3-6 TDF Olivetti 6000 Screen 8

6000 RECEIPT INFORMATION

The following fields control the use of receipts at this terminal.

CUSTOMER BALANCE INFORMATION**1 (N) M**

A code to display balances to the customer for a balance inquiry transaction. For all transactions except balance inquiry transactions, the value in the CUSTOMER BALANCE INFO field on the IDF screen 13 determines what balances are displayed. However, if the transaction is a balance inquiry transaction and the CUSTOMER BALANCE INFO field in the IDF is set to 0 (no information given), the Device Handler process determines the balances to display using this field. The valid values are:

- | | |
|---|------------------|
| 0 | Use both amounts |
| 1 | Use amount 2 |
| 2 | Use amount 3 |

The above amounts are used when BASE24 authorises the transaction using the Positive Balance Authorisation method.

Amount 2 reflects the amount in the AMOUNT ON HOLD/CREDIT BALANCE field on the Positive Balance File (PBF) screen 1 for transactions impacting a credit account. For transactions impacting a debit account, amount 2 reflects the amount in the LEDGER BALANCE/CREDIT LIMIT field on the PBF screen 1.

Amount 3 reflects the amount in the AVAILABLE BALANCE/AVAILABLE CREDIT field on the PBF screen 1 for all transactions.

A change of the setting may require local terminal mask changes to be seen on the Video/Journal.

Default Value	0 (Use both amounts)
Data Name	TDF.MIRR.CUST-BAL-INFO

CUSTOMER BALANCE DISPLAY**1 (N) M**

A code to display customer balance information at this terminal for balance inquiry transactions. For all transactions except balance inquiry transactions, the value in the CUSTOMER BALANCE DISPLAY field on the IDF screen 13 determines how balances are displayed. However, if the transaction is a balance inquiry transaction and the CUSTOMER BALANCE DISPLAY field in the IDF is set to 0 (do not print or display), the Device Handler process determines how to display the balances using this field. The valid values are:

- | | |
|---|--|
| 0 | Display on the screen only |
| 1 | Print on the receipt only |
| 2 | Print on the receipt and display on screen |

A change of the setting may require local terminal mask changes to be seen on the Video/Journal.

Default Value	0 (Display on the screen only)
Data Name	TDF.MIRR.CUST-BAL-DSPY

AUDIT PRINTER OPTION**1 (N) M**

A code indicating whether this terminal has an audit printer. A text description of the code entered is automatically displayed. The valid values are:

2	No audit printer
3	Audit printer

Default Value	2
Data Name	TDF.MIRR.PRT-FLG

LOAD FILENAME**35 (AN)**

The fully expanded Tandem file name used for upline and downline file transfer. If this file name is not specified the Tandem file name specified in the Logical Network Configuration File (LCONF) assign, ATM-DH-OLI6-LOAD-FILE, will be used.

Default Value	No default value
Data Name	TDF.MIRR.LOAD-FNAME

3.6.4 TDF 6000 Screen 9

The Terminal Data File (TDF) Olivetti 6000 screen 9 is displayed for Olivetti 6000 terminals. The DEVICE TYPE field on TDF screen 1 must be set to 32 (Olivetti 6000 Device).

The fields displayed on the TDF Olivetti 6000 screen 9 (Figure 3-7) are explained below.

BASE24-ATM	TERMINAL DATA	REL1 OLIV	96/06/28	16:13	09 OF 12
TERM ID:	FIID: OLIV	REGION:	BRANCH:	LN: REL1	
OLIVETTI 6000 SECURITY INFORMATION					
MASTER KEY:		CHECK DIGITS: 0000			
PIN INFORMATION					
PIN ENCRYPT TYPE: 03 (PIN/PAD (SECURITY DEVICE))					
PIN CHANGE ENCRYPT TYPE: 00 (NO ENCRYPTION)					
LOCAL PIN VERIFY: 1 (DO NOT ALLOW LOCAL PIN VERIFY)					
MESSAGE AUTHENTICATION INFORMATION					
MAC SUPPORT TYPE: 0 (NO MAC SUPPORT)					
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

Figure 3-7 TDF Olivetti 6000 Screen 9

6000 SECURITY INFORMATION

The following fields relate to the master key used by the Device Handler when encrypting the PIN encryption and message authentication transaction keys that it sends to this terminal.

MASTER KEY**16 (AN) M**

The PIN encryption and message authentication master key for this terminal. The valid values are from 0-9 and A-F.

If PINs are transmitted from the terminal encrypted under keys known only to a security module (the PIN ENCRYPT TYPE field is set to 01), this field must contain the terminal master key encrypted by the specific security module being used by the system.

If PINs are transmitted from the terminal in the clear (the PIN ENCRYPT TYPE field is set to 00), this field must be set to zeros.

If the PIN ENCRYPT TYPE field is set to 01, this field must be set to zeros. BASE24-atm uses the terminal master key which is hard coded in the Device Handler process.

If messages between the Device Handler and the terminal are to be authenticated under keys known only to a security module (the MAC SUPPORT TYPE field is set to 3), this field must contain the terminal master key encrypted by or for the specific security module being used by the system.

Default Value	01
Data Name	TDF.MIRR.ENCRYPT-PIN.ENCRYPT-KEYS.M-KEY

CHECK DIGITS**4 (AN)**

The check digits corresponding to the value in the MASTER KEY field for this terminal. The check digits value is obtained from the Hardware Security Module (HSM) at key generation. The valid values for each character in this field are 0-9 and A-F. This field must contain four valid characters.

Default Value	0000
Data Name	TDF.MIRR.ENCRYPT-PIN.ENCRYPT-KEYS.M-KEY-CHK-VALUE

PIN INFORMATION

The following fields relate to the encryption and decryption of data flowing to and from this terminal and to the verification of PIN data presented by this terminal.

PIN ENCRYPT TYPE**2 (N) M**

A code indicating the type of PIN encryption used to secure the PIN when transmitting it across the data communications line. The valid values are:

- | | |
|----|-------------------------------|
| 00 | No Encryption |
| 01 | SIAB (Non-Security Device) |
| 02 | PIN/PAD (Non-Security Device) |
| 03 | PIN/PAD (Security Device) |
| 04 | PIN/PAN (Non-Security Device) |
| 05 | PIN/PAN (Security Device) |

This field is set to:

- | | |
|----|---|
| 00 | if PINs are transmitted from the terminal in the clear. |
| 01 | if PINs are transmitted from the terminal in encrypted SIAB format and decrypted by BASE24 via the DES algorithm. |
| 02 | if PINs are transmitted from the terminal in encrypted PIN/PAD format and decrypted by BASE24 via the DES algorithm. |
| 03 | if PINs are transmitted from the terminal in encrypted PIN/PAD format and under keys that are only in the clear within a security module. |
| 04 | if PINs are transmitted from the terminal in encrypted ANSI standard PIN/PAN format and decrypted by BASE24 via the DES algorithm. |
| 05 | if PINs are transmitted from the terminal in encrypted ANSI standard PIN/PAN format and under keys that are only in the clear within a security module. |

Default Value 01

Data Name TDF.MIRR.ENCRYPT-PIN.PIN-ENCRYPT-TYP

PIN CHANGE ENCRYPT TYPE**2 (N) M**

A code used with a PIN change transaction indicating the type of PIN encryption used to keep the new PIN secure when transmitting it across the data communications line. The valid values are:

00	No Encryption
01	SIAB (Non-Security Device)
02	PIN/PAD (Non-Security Device)
03	PIN/PAD Security Device)
04	PIN/PAN (Non-Security Device)
05	PIN/PAN (Security Device)

The type of encryption performed on the new PIN value must be 00 (clear) or match the value in the PIN ENCRYPT TYPE field as shown below.

PIN ENCRYPT TYPE	PIN CHANGE ENCRYPT TYPE
00	00
01	00 or 01
02	00 or 02
03	00 or 03
04	00 or 04
05	00 or 05

The message B30/b30 PIN Change request/response contains the current and new PIN. If the PIN change request is denied, the customer card will be ejected from the ATM device.

Default Value 00
Data Name TDF.MIRR.PIN-CHG-ENCRYPT-TYP

LOCAL PIN VERIFY**1 (N) M**

A code allowing PIN verification to be performed at this Olivetti ATM device without the involvement of the BASE24 network. The Olivetti ATM device must have PIN verification capability to perform local PIN verification, and the ATM configuration file must be configured for local PIN verification. The local PIN verify flag is used as input to a fraud prevention check for financial and administrative transactions whose PIN has been verified locally by the terminal.

The valid values are:

0	Allow Local PIN Verify
1	Do Not Allow Local PIN Verify (Default)
2	Allow Local PIN Verify for ON-US Transactions Only

Data Name TDF.MIRR.LOCL-PIN-VRFY-FLG

MESSAGE AUTHENTICATION INFORMATION

The following field relates to Message Authentication Code (MAC) support. MAC support provides a method for an institution to validate the contents of transmitted messages, ensuring the integrity of messages received.

MAC SUPPORT TYPE **1 (N)**

A code indicating the type of Message Authentication Code processing (MACing) supported. The valid values are:

- 0 No MAC support (Default)
- 1 Software MAC support
- 3 Hardware MAC support

Data Name TDF.MIRR.MAC.MAC-TYP

4. DEVICE CONTROL PROCEDURES

BASE24-atm provides several ways to control Olivetti 6000 ATMs.

From the Device Control Terminal (DCT) screens, customers can perform all of the functions to control a terminal, including starting and stopping, downline loading, balancing and inquiring the status of the terminal.

Alternatively, commands can be issued from the Network Control Supervisor (NCS) or Network Control Terminal (NCT) screens to stop/start lines and/or stop/start the device handler. Refer to the BASE24 Network Control Operators Manual for more information on the NCS and the NCT screens.

The DCT is the central location of file information on terminals. The DCT module contains all the functions to control the terminal.

4.1 GENERAL PROCEDURES

BASE24 device control and process control functions allow operators to maintain terminals in the BASE24 product family, including Olivetti 6000 ATMs.

The table below outlines the device control and process control functions that should be performed on a regular basis, as well as indicating the DCT screen giving access to the function.

FUNCTION	SCREEN
Warmbooting the Device Handler	DCT screen 4
Signing-on and signing-off terminals	DCT screen 7 or screen 32
Starting and stopping terminals	DCT screen 7
Inquiring the status of terminals	DCT screen 7 or screen 32
Balancing terminals	DCT screen 8 or screen 9
Uploading and downloading files to and from terminals	DCT screen 32
Changing encryption keys used by the terminals	DCT screen 32
Rebooting the terminal	DCT screen 32

For further information on the procedures for warmbooting the Device Handler process (DCT screen 4) and for balancing terminals (DCT screens 8 and 9), refer to the BASE24 Device Control Manual.

The DCT screen 7 and DCT screen 32 are described in this section.

Regular performance of the DCT functions helps to ensure that an institution's network of ATMs operates efficiently and continuously 24 hours a day, seven days a week.

4.1.1 DCT Product Menu

To access the DCT screens, display the DCT Product Menu screen (Figure 4-1), by placing the cursor on the required item on the Virtual Menu (Figure 3-2) and pressing the F1 function key.

With the exception of the DCT Process Control Commands (screen 4), the DCT screens available to authorised operators via the DCT Product Menu depend on:

- The BASE24 product or products purchased by the institution.
- The type or types of terminals the institution operates with the BASE24 product or products.

Figure 4-1 is an example of the DCT Product Menu screen which contains all the DCT options currently available. The DCT Product Menu accessed would not contain all these items.

BASE24-DCT DCT		REL1	96/06/28 15:21 01 OF 02
4	PROCESS CONTROL	7	ATM TERMINAL CONTROL
8	ATM BAL CURRENT CASH	9	ATM BAL STANDARD CASH
10	TELLER TERMINAL CONTROL	15	POS MHI DEVICE LOAD CONTROL
32	ATM OLIVETTI ATM COMMANDS	37	ATM FUJITSU COMMANDS
38	ATM IBM 36XX COMMANDS	39	ATM DIEBOLD COMMANDS
40	ATM DOCUTEL 2300 COMMANDS	41	ATM DOCUTEL ETS 5100 COMMANDS
42	ATM NCR 1770 COMMANDS	43	ATM BURROUGHS RT650 COMMANDS
44	ATM IBM 4730 COMMANDS	45	ATM NCR 5070/5081 COMMANDS
46	ATM OMRON ATM/RCD COMMANDS		
TYPE THE NUMBER OF THE DESIRED FUNCTION IN NEW PAGE FIELD AND KEY F7			
***** BASE24 *****			
NEW PAGE:		FILE DESTINATION: NEW LOGICAL NETWORK ID:	
F7-GOTO F9-NEXT F10-PRINT F11-PAGE-BACK F13-CHANGE-LNET F16-EXIT SF16-LOGOFF			
BLOCK			

Figure 4-1 Example DCT Product Menu

4.2 ATM TERMINAL COMMANDS SCREEN (DCT SCREEN 7)

The ATM Terminal Commands screen (DCT screen 7 - Figure 4-2) allows the operator to perform ATM maintenance functions, regardless of the type of ATM. Maintenance functions and commands are detailed in section 4.2.3.

This section contains:

- DCT screen 7 field descriptions
- DCT screen 7 function keys
- DCT screen 7 commands

Closing an ATM for balancing is detailed in the sections describing the ATM Balance Current Cash (DCT screen 8) and the ATM Balance Standard Cash (DCT screen 9) procedures. Refer to the appropriate sections in the BASE24 Device Control Manual for more details.

4.2.1 DCT Screen 7 Field Descriptions

The ATM Terminal Commands DCT screen 7 (Figure 4-2) allows the operator to perform various functions on all ATMs defined in the network Terminal Data File (TDF) regardless of the terminal type. These functions include inquiring on the status of terminals and stations, starting and stopping stations, and opening and closing ATMs.

When the function key F2 (STATUS) is pressed, the terminal operable status description will be displayed. This information consists of:

- Header text
- TO status line
- Number of inactivity timeouts for a particular Olivetti 6000 ATM device, if greater than zero
- General device status details.

This is followed by a Device Status report consisting of up to eight lines of peripheral error conditions.

The fields displayed on the DCT screen 7 are explained below.

BASE24-DCT		TERMINAL COMMANDS	REL 1	94/08/19	19:06	07 OF 46
ATM TERMID:		FIID:		REGION:		BRANCH:
STATUS:	4730	CONSOLE ID:				
***** BASE24 ***** NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID: F2-STATUS F3-OPEN F5-CLOSE F6-STR F7-GOTO F8-STOP F9-STATUS ALL F12-HELP						

Figure 4-2 DCT Screen 7

ATM TERMID**16 (AN) M**

The ATM terminal identifier that specifies the ATM on which commands entered on this screen will be performed. This identifier must have a record in the Terminal Data File (TDF).

Default Value No default value

FIID**4 (AN)**

The FIID of the financial institution that owns the ATM terminal. The FIID entered in this field must be defined in the Institution Definition File (IDF) or ***** (asterisks).

Default Value ***** (asterisks)

REGION**4 (AN)**

The region in which the specified ATM is located. The region is determined by the institution identified in the FIID field, if applicable.

Default Value ***** (asterisks)

BRANCH**4 (AN)**

The branch in which the specified ATM is located. The branch is determined by the institution identified in the FIID field and is a subdivision of the region displayed in the REGION field.

The financial institution assigns terminals to a region and branch using fields in the Terminal Data File (TDF). The values in the REGION and BRANCH fields must match the region and branch assigned to the terminal in the TDF.

Default Value ***** (asterisks)

STATUS**7 (AN)**

The type of terminals for which status information is displayed via the STATUS ALL function. The valid values are:

ALL	All terminals
BALANCE	Terminals that are balanced
CLOSED	Terminals that are closed
DOWN	Terminals that are down
FAULT	Terminals that are faulted
OLI6	Olivetti ATM terminals
NOT UP	Terminals that are down, closed or faulted
NOTRAN	Terminals with no transactions in progress
OPEN	Terminals that are open
TRAN	Terminals with transactions in progress
UNBAL	Terminals that are unbalanced
UP	Terminals that are up
Default Value	No default value

4730 CONSOLE ID**1 (N)**

Not required for Olivetti 6000 ATMs.

4.2.2 DCT Screen 7 Function Keys

The usage of the function keys assigned for the ATM Terminal Commands screen (DCT screen 7) is explained in the following table. The first column shows the function key. The second and third columns provide a description of how the function key works for the screen.

KEYS	FUNCTION	DESCRIPTION
F1	VALIDATE	Checks selected fields for invalid entries. If an entry is invalid, this function highlights the field and displays an error message indicating the information that needs to be changed or added.
F2	STATUS OF TERMINAL	Displays status information for the specified ATM components that are faulted. A number preceding the fault message indicates the number of occurrences that the fault has been reported.
F3	OPEN TERMINAL	Opens the specified terminal.
F4	STATION INQUIRY	Displays status information for the specified station.
F5	CLOSE	Closes the specified terminal.
F6	START	Starts the specified terminal.
F7	GOTO NEW SCREEN	Displays the DCT screen corresponding to the number entered in the NEW PAGE field.
F8	STOP	Stops the specified terminal.
F9	STATUS ALL	Displays the status of all terminals of the type specified in the STATUS field that are owned by the specified FIID, region and branch of the financial institution.
F10	PRINT	Sends the image of the DCT screen 7 to the spooler location specified on the Logon screen.
F12	DISPLAY HELP SCREEN	Displays function key information or menu options.
F16	EXIT OR GOTO FILE	Exits from the DCT screen 7 to the DCT Product Menu screen, if no entry is specified in the NEW LOGICAL NETWORK ID or FILE DESTINATION fields. If values are entered in these fields, this function displays the name of the specified file in the specified logical network.
SF16	LOGOFF	Exits from BASE24-atm and displays the BASE24 Logon screen.

4.2.3 DCT Screen 7 Commands

The following commands are routinely executed from the ATM Terminal Commands DCT screen 7:

- Inquiring on the status of a terminal
- Inquiring on the status of all terminals
- Inquiring on the status of a station
- Opening/Starting an ATM
- Closing/Stopping an ATM
- Starting a Station
- Stopping a Station

The procedures for executing the above commands are briefly described below.

1. **Inquiring on the status of a terminal**—an inquiry can be made on the status of a particular terminal. In response to the inquiry, the system displays a detailed status report for the specified device. If the ATM is deemed to be HUNG this will be indicated in the status report.
2. **Inquiring on the status of all terminals**—an inquiry can be made on the status of all terminals, or all terminals owned by a specified financial institution and/or of a specified type. In response to the inquiry, the system displays a summary status for each of the specified devices, including:

⇒ type of device

⇒ physical status

⇒ logical state

⇒ an indication if the ATM is currently reporting faults.

If any ATMs are deemed to be HUNG this will be indicated in the status report.

When a status inquiry is made from the DCT screen 7, the system interrogates the Terminal Data File (TDF) for the last reported status of the devices and displays the information. It does not interrogate the ATMs directly.

However, if a status inquiry is made from the Olivetti 6000 DCT screen 32, the system sends a status inquiry to the ATM to determine its current status. The status report is logged to the network logging device.

Therefore, a DCT screen 7 status inquiry displays the results of the inquiry immediately, but that report is the last reported status, whilst a DCT screen 32 status inquiry obtains the current status directly from the ATM, but it is necessary to refer to the network log for the results of this report.

3. **Inquiring on the status of a station**—an inquiry can be made on the status of a particular station. In response to the inquiry, the system displays a detailed status report for the specified station.
4. **Opening/Starting an ATM**—when an ATM is opened (the value of the STATE field is OPEN), it displays a Welcome screen for initiating transactions using EFT cards.

Before any messages can be sent to or received from an ATM, the ATM station must be started and the value in the STAT field is Up. Starting a station initiates polling to the ATM and allows the ATM to be signed-on. If the ATM is the first station on the communications line, the line is also started.

5. **Closing/Stopping an ATM**—when an ATM is closed (the value of the STATE field is CLOSED), it displays a Temporarily Out of Order message and prevents the initiating of transactions using EFT cards. When transactions in progress at the time the CLOSE is issued are concluded, a corresponding message is sent to the log.
6. **Starting a Station**—before any messages can be sent to or received from an ATM, the communications line and the ATM station must be started. Starting a station initiates polling to the ATM and also opens the ATM.

When the first station on a communications line is started, the line is also started.

7. **Stopping a Station**—when a station is stopped, polling stops and XPNET considers the station offline. After the ATM timeout interval (usually 60 seconds) has expired, an Out of Service message is displayed on the ATM screen.

If a customer transaction is in progress when a station is stopped, the transaction is not completed. When the station is restarted, the transaction-in-progress is reversed. When the last station being polled on a line is stopped, the communications line is stopped.

To shutdown an ATM, close the ATM before stopping the station.

For a detailed description of the procedures to execute the commands from the DCT screen 7, refer to the BASE24 Device Control Manual.

4.3 OLIVETTI 6000 ATM COMMANDS SCREEN (DCT SCREEN 32)

The Olivetti 6000 ATM Commands screen (DCT screen 32 - Figure 4-3) allows the operator to perform Olivetti 6000 ATM maintenance functions. Maintenance functions include opening (signon), closing (signoff) and resetting the specified terminal, as well as obtaining terminal status and supplying status information.

The File Transfer Service (FTS) is initiated from the Device Control Table (DCT). The FTS provides three options to transfer files in compressed format:

- **Upload Directory**

The Olivetti 6000 ATM device uploads all the file names in the requested directory specified in the LOAD FILE NAME PREFIX field from the specified ATM to BASE24. The Device Handler logs all the file names to the Event Log. There is no actual transfer of files.

- **Upload File**

The ATM device uploads the requested file to BASE24. The Device Handler stores this file in an unstructured Tandem disc file which is specified in the Terminal Data File (TDF) or in an LCONF assign if the TDF file name is spaces.

- **Download File**

BASE24 downloads an unstructured Tandem disc file to the Olivetti ATM device. The disc file is specified in the Terminal Data File (TDF) or in an LCONF assign if the TDF file name is spaces.

The FTS can upload a new file from an Olivetti ATM device (or a PC emulating a device) to BASE24 and then download this file to each ATM device requiring the new file. The maximum file size that can be transferred is 999,999 bytes. Configuring the maximum Data Block Size (DBS) to a large value can have adverse performance implications for transaction processing. It is recommended that a maximum value of 1024 bytes be used.

If a PC is used to generate files to be distributed via BASE24, it is recommended that MACing be switched off in the TDF record for this ATM device. Otherwise, the PC will need a software MACing facility or a special DEM board for hardware MACing.

If a bad MAC is detected, the Device Handler and the Olivetti ATM device (or a PC emulating a device) will exit the FTS to allow normal transactions to resume (assuming that the ATM device is signed-on).

This section contains:

- DCT screen 32 field descriptions
- DCT screen 32 function keys
- DCT screen 32 commands:
 - ⇒ Download a key to an ATM (LOADKEY)
 - ⇒ Upload a directory listing from an ATM (UPLOAD DIR)
 - ⇒ Upload a file from an ATM (UPLOAD FILE)
 - ⇒ Download a file to an ATM (DOWNLOAD)
 - ⇒ Reboot the ATM controller (REBOOT)
 - ⇒ Set Date and Time (DATETIME)

4.3.1 DCT Screen 32 Field Descriptions

The Olivetti 6000 ATM Commands screen (DCT screen 32 - Figure 4-3) allows the operator to perform various functions on Olivetti 6000 ATM terminals, including loading a new communication key to an ATM and upline and downline loading of ATM files.

The fields displayed on the DCT screen 32 are explained below.

BASE24-DCT		OLIVETTI 6000 ATM		REL1	OLIV	96/06/28	16:37	32 OF 46
ATM TERMINAL ID: _____		FIID: _____						
LOAD KEY: -								
LOAD FILE NAME PREFIX: _____								
SELECT ONE AND PRESS F1								
-	LOADKEY							
-	UPLOAD DIR							
-	UPLOAD FILE							
-	DOWNLOAD							
-	REBOOT							
-	DATETIME							
***** BASE24 *****								
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:				
F1-SELECT F2-STATUS F3-SIGNON F5-SIGNOFF F6-SIGNOFF (IMM) F9-RESET F16-EXIT								

Figure 4-3 DCT Screen 32

ATM TERMINAL ID**16 (AN) M**

The ATM terminal identifier that specifies the Olivetti 6000 ATM to which commands entered on this screen are directed. This identifier must have a record in the Terminal Data File (TDF).

Default Value No default value

FIID**4 (AN)**

The FIID of the financial institution that owns the ATM terminal. The FIID entered in this field must be defined in the Institution Definition File (IDF) or **** (asterisks).

Default Value **** (asterisks)

LOAD KEY**1 (N) M**

The type of key downloaded to the specified ATM when the LOADKEY command is executed. The valid values are:

- 0 Master key
- 1 Communication key
- 2 PIN key
- 3 MAC key

Default Value One space

LOAD FILE NAME PREFIX**34 (AN) M**

The name of a file or directory to be uploaded to or downloaded from the ATM.

Default Value Blanks

LOADKEY**(Cursor placement only)**

Placing the cursor in this field and pressing the F1 (SELECT) function key loads the key type specified in the LOAD KEY field to the specified ATM.

UPLOAD DIR**(Cursor placement only)**

Placing the cursor in this field and pressing the F1 (SELECT) function key uploads the listing of the directory specified in the LOAD FILE NAME PREFIX field from the specified ATM.

UPLOAD FILE**(Cursor placement only)**

Placing the cursor in this field and pressing the F1 (SELECT) function key uploads the file specified in the LOAD FILE NAME PREFIX field from the specified ATM.

DOWNLOAD**(Cursor placement only)**

Placing the cursor in this field and pressing the F1 (SELECT) function key downloads the file specified in the LOAD FILE NAME PREFIX field to the specified ATM.

REBOOT**(Cursor placement only)**

Placing the cursor in this field and pressing the F1 (SELECT) function key causes the ATM controller to reboot.

DATETIME**(Cursor placement only)**

Placing the cursor in this field and pressing the F1 (SELECT) function key causes the Device Handler to remotely set the local date and time of the Olivetti 6000 PC clock; for example, in the case of daylight savings.

The DATETIME command can also be issued from the Network Control Supervisor (NCS) and multiple Olivetti devices sent via an NCS READ file. This enables the operator to submit a single command to update all devices in the network.

The format of the NCS command to load the date and time is:

DELIVER <processname>, '9500DATETIME <terminalidentifier>'

4.3.2 DCT Screen 32 Function Keys

The usage of the function keys assigned for the Olivetti 6000 ATM Commands screen (DCT screen 32) is explained in the following table. The first column shows the function key. The second and third columns provide a description of how the function key works for the screen.

KEYS	FUNCTION	DESCRIPTION
F1	SELECT	Checks selected fields on the screen for valid entries and, if these entries are valid, executes the selected command. If an entry is invalid, this function highlights the field and displays an error message indicating the information that needs to be changed or added.
F2	STATUS	Displays status information for the specified terminal on the network logging facility.
F3	SIGNON	Opens the specified ATM.
F5	SIGNOFF	Closes the specified ATM on completion of the transaction in progress.
F7	GOTO	Displays the DCT screen corresponding to the number entered in the NEW PAGE field.
F9	RESET	Resets the terminal transaction and load status.
F10	PRINT	Sends the image of the DCT screen 32 to the spooler location specified on the Logon screen.
F12	DISPLAY HELP SCREEN	Displays function key information or menu options.
F16	EXIT OR GOTO FILE	Exits from the DCT screen 32 to the DCT Product Menu screen, if no entry is specified in the NEW LOGICAL NETWORK ID or FILE DESTINATION fields. If values are entered in these fields, this function displays the name of the specified file in the specified logical network.
SF16	LOGOFF	Exits from BASE24-atm and displays the BASE24 Logon screen.

4.3.3 DCT Screen 32 Commands

The following commands are routinely executed from the Olivetti 6000 ATM Commands DCT screen 32:

- Download a key to an ATM (LOADKEY)
- Upload a directory listing from an ATM (UPLOAD DIR)
- Upload a file from an ATM (UPLOAD FILE)
- Download a file to an ATM (DOWNLOAD)
- Reboot the ATM controller (REBOOT)
- Set Date and Time (DATETIME)

The procedures for executing the above commands are described below.

4.3.3.1 Download a Key to an ATM (LOADKEY)

The LOADKEY command downloads a new master key, communication key or PIN key to the specified ATM. To download a key to an ATM:

1. Enter a valid Olivetti 6000 ATM identifier in the ATM TERMINAL ID field.
2. Enter the financial institution identifier that owns the ATM in the FIID field.
3. Enter the key type to be downloaded in the LOADKEY field. The valid values are:

0	Master key
1	Communication key
2	Load PIN key
3	Load MAC key

If the TDF Olivetti 6000 screen 9 MAC SUPPORT TYPE field is set to 3 (Hardware MAC Support) or 1 (Software MAC Support), the LOADKEY field value of 3 (Load MAC key) is permitted. However, if the MAC SUPPORT TYPE field is not set to 3 or 1, the LOADKEY field cannot be set to 3.

4. Place the cursor in the LOADKEY field.
5. Press the F1 (SELECT) function key.
6. Check the screen to ensure the message Command Sent to Network is displayed. If not, repeat steps 1-5.
7. Check the log for messages indicating that the key has been loaded successfully.

4.3.3.2 Upload a Directory Listing from an ATM (UPLOAD DIR)

The UPLOAD DIR command uploads a listing of the directory specified in the LOAD FILE NAME PREFIX field from the specified ATM and displays the file names in that directory on the log. To upload a directory listing from an ATM:

1. Enter a valid Olivetti 6000 ATM identifier in the ATM TERMINAL ID field.
2. Enter the financial institution identifier that owns the ATM in the FIID field. Alternatively, enter ***** (asterisks).
3. Enter the name of the DOS directory to be uploaded from the specified ATM in the LOAD FILE NAME PREFIX field.
4. Place the cursor in the UPLOAD DIR field.
5. Press the F1 (SELECT) function key.
6. Check the screen to ensure the message Command Sent to Network is displayed. If not, repeat steps 1-5.
7. Check the log to view the names of all files located in the specified directory.

4.3.3.3 Upload a File from an ATM (UPLOAD FILE)

The UPLOAD FILE command uploads the file specified in LOAD FILE NAME PREFIX field from the specified ATM. To upload a file from an ATM:

1. Enter a valid Olivetti 6000 ATM identifier in the ATM TERMINAL ID field.
2. Enter the financial institution identifier that owns the ATM in the FIID field. Alternatively, enter ***** (asterisks).
3. Enter the name of the file to be uploaded from the specified ATM in the LOAD FILE NAME PREFIX field.
4. Place the cursor in the UPLOAD FILE field.
5. Press the F1 (SELECT) function key.
6. Check the screen to ensure the message Command Sent to Network is displayed. If not, repeat steps 1-5.
7. Check the log for messages indicating that the specified file has been uploaded successfully. The file will be uploaded to the Tandem file name specified in the LOAD FILENAME field on the Terminal Data File (TDF) screen 8. If this field is blank or the file is not found, the Olivetti Device Handler process defaults to the file name specified in the Logical Network Configuration File (LCONF) assign, ATM-DH-OLI6-LOAD-FILE.

4.3.3.4 Download a File to an ATM (DOWNLOAD)

The DOWNLOAD command downloads a file from the Tandem to the ATM. The file will have previously been uploaded to the Tandem using the DCT UPLOAD command. The file will be downloaded to the file location specified in the LOAD FILENAME PREFIX field. To download a file to an ATM:

1. Enter a valid Olivetti 6000 ATM identifier in the ATM TERMINAL ID field.
2. Enter the financial institution identifier that owns the ATM in the FIID field. Alternatively, enter ***** (asterisks).
3. Enter the destination file location for the ATM in the LOAD FILE NAME PREFIX field.
4. Place the cursor in the DOWNLOAD field.
5. Press the F1 (SELECT) function key.
6. Check the screen to ensure the message Command Sent to Network is displayed. If not, repeat steps 1-5.
7. Check the log for messages indicating that the specified file has been downloaded successfully. The file will be downloaded from the Tandem file name specified in the LOAD FILENAME field on the Terminal Data File (TDF) screen 8. If this field is blank or the file is not found, the Olivetti Device Handler process defaults to the file name specified in the Logical Network Configuration File (LCONF) assign, ATM-DH-OLI6-LOAD-FILE.

4.3.3.5 Reboot the ATM (REBOOT)

The REBOOT command issues the ENTER FILE TRANSFER SERVICE command to the Olivetti 6000 Device Handler to prepare the ATM for a reboot. The ATM responds with the FILE TRANSFER SERVICE command request to indicate that it is rebooting. Reboot resets the device PC controller, communicator board and peripherals. To reboot the ATM:

1. Enter a valid Olivetti 6000 ATM identifier in the ATM TERMINAL ID field.
2. Enter the financial institution identifier that owns the ATM in the FIID field. Alternatively, enter ***** (asterisks).
3. Place the cursor in the REBOOT field.
4. Press the F1 (SELECT) function key.
5. Check the screen to ensure the message Command Sent to Network is displayed. If not, repeat steps 1-4.
6. Check the log for messages indicating that the reboot is in progress.

5. SUPPORTED TRANSACTIONS

The Olivetti 6000 Device Handler supports a number of customer and administrative transactions for the Olivetti 6000 ATMs.

5.1 CUSTOMER TRANSACTIONS

The customer transactions supported by the Olivetti 6000 Device Handler are:

- Deposit (Envelope and Commercial)
- Electronic Payment - only supported if multi-account cards (card review) is configured
- Inquiry
- LOGONLY
- Message Enclosed
- Payment Enclosed
- PIN Change
- Statement Print
- Transfer - only supported if multi-account cards (card review) is configured
- Withdrawal
- Currency Exchange.

5.1.1 Account Mapping

For BASE24 to support the above customer transactions, the Olivetti 6000 ATM is configured to map the ATM account types to the BASE24 account types as detailed in the table below.

ATM ACCOUNT TYPES		BASE24 ACCOUNT TYPES	
02	Credit Card	31	Credit Card
04	Demand Deposit	01	Checking
05	Time Deposit	11	Savings

5.1.2 Card Capture

The Olivetti 6000 Device Handler supports card capture, for Olivetti 6000 ATMs, by forwarding a card capture code in reply to certain financial transaction messages. When a card is captured as a result of this code, the ATM, if appropriately configured, produces a confirmation slip for the customer. The text of the confirmation slip is configured at the ATM.

The card capture code is sent as a one byte field in the following messages:

- b01 Card review response
- b02 Deposit authorisation response
- b03 Debit authorisation response
- b05 Transfer authorisation response
- b11 Balance authorisation response
- b14 Instant statement response
- b15 Electronic payment authorisation response
- b16 Payment enclosed authorisation response
- b21 Message to financial institution authorisation response
- b30 PIN change response.

The card capture code will not be sent in the following message:

- b09 Reversal authorisation response.

5.2 ADMINISTRATIVE TRANSACTIONS

The administrative transactions supported by the Olivetti 6000 Device Handler are:

- Cartridge Maintenance
- Cutoff Request (Close Period).

Depending on the configuration of the ATM processing control parameters, terminal balancing is handled in one of the following ways:

- **Within Window Balancing**—terminals are balanced once per day in a specified period.
- **No Window Balancing**—terminals are balanced at any time and more than once a day.

The following points should be observed when balancing terminals and replenishing cash.

5.2.1 Cartridge Maintenance

During a cartridge maintenance operation, a message is sent to BASE24. No window or date restrictions exist for this type of operation. The BASE24 Terminal Data File (TDF) cash increase/decrease totals (TDF screen 4) reflect the number of bills added/removed by the service operator, as reported on the ATM back panel.

5.2.2 Cutoff Request (Close Period)

When the ATM service operator initiates a cutoff request for terminal balancing, this resets the ATM's counters and sends a cutoff request to BASE24. By the time BASE24 receives the cutoff request, the terminal is already out of service. Therefore, BASE24 cannot deny the balance or cash increase. Accordingly, careful balancing procedures must be followed.

BASE24 totals reflect the cutoff balance and will be reset to the number of bills reported by the service operator, on the ATM back panel, provided that:

- The transaction occurs in the normal window.
- One-time window balancing has been configured.
- The ATM has not already been balanced in the current window.

If the transaction occurs under one of the following circumstances:

- One-time window balancing has been configured.
- The transaction occurs outside the normal window.
- The terminal has already been balanced during the current window.

the following message:

```
629      Station <station identifier>; cut-off request invalid; out of  
        balance situation present
```

will be logged by BASE24. This message indicates a possible out-of-balance situation between the Terminal Data File (TDF) totals and the ATM totals.

Accordingly, with one-time window balancing set, it is recommended that:

- The ATM is balanced only once a day in the normal window.
- The cash is replenished with a full cartridge (so that the service operator does not need to calculate the number of bills present at the ATM and, hence, possibly introduce an error).

5.3 STATUS MESSAGES

The status messages available from BASE24-atm include:

- **Terminal Totals**

Provides the device with transaction totals. It is used during back panel operations on the ATM.

- **Terminal Status**

Provides reports on the overall status of the device. It is used when STATUS is requested from the DCT screen 32 (see section 4.3) and when the device provides an unsolicited status message.

The terminal status message is used by the Device Handler to poll the ATM. The Device Handler initiates a terminal status request if no messages are received from the ATM in the time interval. This interval is specified in the LCONF parameter ATM-DH-OLI6-POLL-INTERVAL.

⇒ If no message is received during this interval the Device Handler sends a terminal status request to the ATM.

⇒ If a message is received from the ATM during the response time interval, the poll request is deemed to have been responded to. This interval is defined by the LCONF parameter ATM-DH-OLI6-CMD-TIMEOUT.

⇒ If no message is received during the response time interval, the Device Handler sends a further terminal status request after a retry time interval. This interval is specified in the LCONF parameter ATM-DH-OLI6-RETRY-INTERVAL. The Device Handler also logs the message 1031 to the BASE24 event log and increments a count of the number of retries sent. If a response is received during the response time interval the retry counter is reset to zero.

⇒ The Device Handler retries up to the maximum number of retries allowed, as specified in the LCONF parameter ATM-DH-OLI6-POLL-RETRY-MAX. Once the maximum number is exceeded, the Device Handler logs the message 1032 to the BASE24 event log and marks the ATM as HUNG.

⇒ If a message is received from the ATM after it has been marked as HUNG, the Device Handler marks the ATM as NOT HUNG and resets the retry counter to zero.

Retry processing is performed for terminal status requests initiated from the DCT screen 32.

- **Peripheral Status**

Provides reports on the peripheral status. It is used when the device provides an unsolicited peripheral status message.

- **Cassette Status**

Provides information to BASE24 on any cash increases/decreases or cutover at the ATM during back panel operations.

- **Currency Exchange Totals**

Provides the device with current transaction totals since the last currency exchange transaction request. The Device Handler logs a message to the BASE24 event log if the LCONF parameter ATM-DH-OLI-LOG-CE-ACTIVITY has been set to Log A06 messages.

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6. TRANSACTION SECURITY

This section describes the transaction security features supported by the Olivetti 6000 Device Handler.

6.1 PIN VERIFICATION

The Olivetti 6000 Device Handler supports Atalla or Racal Hardware Security Modules (HSMs).

If a security device is used for PIN verification, the master key for the terminal must remain static. Any requests from the Olivetti 6000 specific DCT screen 32 to load the master key for terminals using a security device will be disallowed by the Olivetti 6000 Device Handler.

Message encryption is only applicable to the terminals where the PIN ENCRYPT TYPE field is set to 01 (Hardware Security Module Not Present).

For more details on firmware versions, refer to the Appendix B.

6.1.1 Local PIN Check

The Olivetti 6000 Device Handler supports PIN check at the Olivetti 6000 ATM device. The Card Processing Code (CPC) values of 1, 3, 5 and 9 indicate local PIN check.

The Device Handler checks the CPC values and sets the Standard Internal Message (STM) fields PIN tries to Z and the PIN block to ASCII zeros. The Device Handler formats the value of the LOCAL PIN VERIFY field on the Terminal Data File (TDF) Olivetti 6000 screen 9 to the AT50-TKN and includes this token in the request to the Authorisation process.

The Authorisation process checks the existence of the AT50-TKN token. If this token is present, the Authorisation process will perform fraud processing. The transaction will be processed if the value in the LOCAL PIN VERIFY field is valid; otherwise, the value F is returned in the ERR FLG field of the AT50-TKN token. A response will be returned to the device. If the token is not present, the Authorisation process will process the transaction, but without fraud processing.

For more details on BASE24 tokens, refer to the BASE24 Tokens Manual.

6.1.2 PIN Block Formats

The Olivetti 6000 Device Handler supports the ANSI PIN/PAN PIN block format and the PIN/PAD PIN block. A proprietary PIN block format known as SIAB is also supported.

On the Terminal Data File (TDF) Olivetti 6000 screen 9, the PIN ENCRYPT TYPE field indicates which PIN block format is being used by the given ATM:

- 00 No Encryption
 PINs are transmitted from the terminal in the clear.
- 01 SIAB (Non-Security Device)
 PINs are transmitted from the terminal in encrypted SIAB format and decrypted via the DES algorithm.
- 02 PIN/PAD (Non-Security Device)
 PINs are transmitted from the terminal in encrypted PIN/PAD format and decrypted by BASE24 via the DES algorithm.
- 03 PIN/PAD (Security Device)
 PINs are transmitted from the terminal in encrypted PIN/PAD format and under keys that are only in the clear within a security module.
- 04 PIN/PAN (Non-Security Device)
 PINs are transmitted from the terminal in encrypted ANSI x9.8 standard PIN/PAN format and decrypted by BASE24 via the DES algorithm.
- 05 PIN/PAN (Security Device)
 PINs are transmitted from the terminal in encrypted ANSI x9.8 standard PIN/PAN format and under keys that are only in the clear within a security module.

The TDF screen 9 PIN CHANGE ENCRYPT TYPE field is used when the PIN Change transaction is allowed. It indicates whether the new PIN is to be encrypted or in the clear. This field corresponds to the PIN ENCRYPT TYPE field for allowable values.

For encryption of the PIN/PAD or PIN/PAN PIN blocks, the Olivetti 6000 ATM device uses a PIN block key that is downloaded by the Device Handler.

For hardware encryption, the PIN block key is generated by the HSM. For software encryption, the PIN block key is generated in the software. The PIN block key is encoded in the terminal MASTER KEY field for transmission to the Olivetti 6000 ATM device.

6.2 MACing SUPPORT

Full or partial Message Authentication Code processing (MACing) is supported by Olivetti 6000 devices.

For partial MACing, any part of an identified message can be selected. The selection must be for a specified length and up to ten fields per message.

The MAC SUPPORT TYPE field on the Terminal Data File (TDF) Olivetti 6000 screen 9 specifies whether software MACing or hardware MACing is supported by the terminal. The MAC key is loaded when:

- A MAC key request is received from the Olivetti 6000 ATM device as part of the signon procedure.
- The count of consecutive bad MAC messages has surpassed a configurable limit.
- The LOADKEY command is executed from the DCT screen 32.

The number of consecutive bad MACs allowed is specified in the Logical Network Configuration File (LCONF). If this threshold is reached, the Device Handler automatically loads a new MAC key. If a subsequent bad MAC is received by the Device Handler, the Olivetti 6000 ATM device will be automatically taken out of service by a Signoff Immediately command (z22).

The sequence for the loading of a new MAC key is:

- When the number of bad MACs limit is reached, send a z0L (Enter dynamic key change service) message to the Olivetti 6000 ATM device.
- Proceed with the dynamic key change loading the new MAC key.

If the load is successful and the first transaction after the load is verified successfully, the bad MAC counter is reset. If the load is successful and the first transaction after the reload fails the MAC verification, then the 6000 ATM device is signed off.

The Device Handler supports MACing for all message types from/to the 6000 ATM device with the exception of the following:

- ⇒ A05
- ⇒ A0R
- ⇒ Dxx/dxx
- ⇒ Sxx/sxx
- ⇒ Zxx/zxx

The MAC field is not present in the above messages. For a comprehensive list of the types of messages supported, refer to the Appendix A.

The DHNAMES MAC Selection Table is configurable and contains the message types as required. Each entry in the table consists of the message type and ten offset-length pairs. This table is sourced by the Device Handler at compile time. Modifications made will require a Device Handler recompile.

For financial messages (Bxx/bxx), with the exception of reversals (B09/b09), the message type in the DHNAMES MAC Selection Table includes the value of the Card Processing Code (CPC). For file transfer type messages (G01/g01), the message type includes the value of the Service Type (ST).

All table offsets are taken from 1. Field entries in the table that are not used have offset-length values set to -1. The table is terminated by a null value. All messages in the table are MACed. Full MACing is performed when the first offset entry for a particular message is -1. Partial MACing is performed when the offsets and lengths for a particular message have values other than -1. The 6000 ATM device contains another similar configurable table. These two tables must match for MACs to be successfully verified and generated.

When MACing is selected, the MAC value resides in the last 8 bytes of a message. If a particular message is not found in the DHNAMES MAC Selection Table, the MAC field is set to ASCII zeros. The exceptions are A05, A0R, Sxx/sxx, Dxx/dxx and Zxx/zxx type messages where the MAC field will never be present.

When MACing is not selected, the MAC field is not present in any message to the device.

6.3 DYNAMIC KEY CHANGE

The PIN Block and MAC keys are changed automatically on a regular basis.

The Logical Network Configuration File (LCONF) param, ATM-DH-OLI6-KEY-CHG-TIME, specifies the time of day at which the Device Handler will initiate a dynamic key change. Following the first approved withdrawal after the specified time, the Device Handler will initiate the dynamic key change.

A new PIN block key and a new MAC key will be loaded. However, if MACing is not configured, only the PIN block key will be loaded.

The down-time is minimal and the ATM device is automatically re-opened for use.

To change the dynamic key change time after the next dynamic key change has taken place, perform a warmboot. The Device Handler reads the param but the timer will not be reinitialised. The new time will be effective when the current timer has expired.

In addition, new keys are loaded to the device as part of every signon sequence.

7. ATM PRINTER SUPPORT

For the standard Olivetti 6000 Device Handler, the receipts printed by the ATM comprise two sections:

- The first section consists of all but the last line of the receipt and is formatted and controlled by the ATM. The account balance and the account number (if a mono account transaction) are returned by BASE24 to the ATM. However, the formatting of this portion of the receipt is controlled by the ATM.

For more information on receipt formatting, refer to the OLIVETTI/BASE24 Connection Getting Started/Personalisation Guide (Release 4.0).

- The second section consists of the last line of the receipt and is formatted by the Olivetti 6000 Device Handler. This line contains the transaction sequence number and the customer card number. The format of this line is specified from the 6000 verbiage tables in the BASE24 file DHNAMES. This file contains customised ATM data of all ATM types.

Figure 7-1 provides an example of an ATM receipt.

**** 6000 APPLICATION ****	)	
	)	
** BALANCE **	)	
01/06/95	)	
	)	Controlled by the ATM
FII BRANCH : 007700770 0000	)	
ACCOUNT NUMBER : 161616	)	
CURRENT BALANCE : 840 + 0000009,620.00	)	
AVAILABLE BALANCE : 840 + 0000009,620.00	)	
RECORD NO. 22 CARD NO.	)	Controlled by the Olivetti 6000
22772277131313130	)	Device Handler
*** THANK YOU ***	)	

Figure 7-1 Example of an ATM Receipt

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A. MESSAGE TYPES

The table below lists the messages used in communication between the Olivetti 6000 Device Handler and Olivetti 6000 ATMs.

MESSAGE		TRANSACTION
A02/a02	Current terminal totals request/response	Status
A05	Terminal overall status report	Status
A06/a06	Currency exchange request/response	Status
A0Q/a0Q	Cash cartridge maintenance request/response	Administrative
A0R/a0R	Terminal peripheral status request/response	Status
B01/b01	Card review request/response	Customer
B02/b02	Deposit authorisation request/response	Customer
B03/b03	Debit authorisation request/response	Customer
B05/b05	Transfer authorisation request/response	Customer
B09/b09	Reversal authorisation request/response	Customer
B11/b11	Balance authorisation request/response	Customer
B14/b14	Instant statement request/response	Customer
B15/b15	Electronic payment authorisation request/response	Customer
B16/b16	Payment enclosed authorisation request/response	Customer
B21/b21	Message to financial institution authorisation request/response	Customer
B30/b30	PIN change request/response	Customer
D0B/d0B	Signoff request/response	Administrative
D0C/d0C	Global information exchange request/response (includes date/time and terminal definition)	Administrative
D0L/d0L	Set field length request/response (optional message)	Administrative
G00/g00	FILE TRANSFER SERVICE (FTS) command request/response	Administrative
G01/g01	FTS data transfer request/response	Administrative
S00/s00	Next service request/response — SECURITY DATA MAINTENANCE SERVICE (SDMS) command	Administrative
S01/s01	Load communication (SDMS) key request/response	Administrative
S02/s02	Dynamic key change request/response	Administrative

MESSAGE		TRANSACTION
z00	Positive status poll	Administrative
z01	Unsolicited request	Status
z02	Signon if possible	Administrative
z12	Signoff after current transaction	Administrative
z22	Signoff immediately	Administrative
z0G	Enter file transfer status	Administrative
z0L	Enter dynamic key change service	Administrative
z0S	Enter SDMS	Administrative

Below is an example of a signon message sequence. The 6000 ATM device checks the firmware version sent by the Olivetti Device Handler. The signon sequence will fail if the value is invalid.

SIGNON MESSAGE SEQUENCE			
		⇐	z02 Signon if possible
D0C	Global information exchange request	⇒	
		⇐	d0C Global information exchange response
D0L	Set field length request (optional message)	⇒	
		⇐	d0L Set field length response
S01	Load communication (SDMS) key request (optional message)	⇒	
		⇐	s01 Load communication (SDMS) key response
S00	Next service (SDMS command) request (optional message)	⇒	
		⇐	s00 Exit SDMS command reply
A05	Terminal overall status report	⇒	

B. ATM FIRMWARE

The Olivetti 6000 Device Handler supports Olivetti 6000 ATMs with the following firmware versions:

- Version A
- Version B
- Version C

This guide documents devices which support Version C only.

The ATM firmware version is specified in the VERSION ID field in the Terminal Data File (TDF) Olivetti 6000 screen 7. The Olivetti 6000 ATM device checks the firmware version sent by the Olivetti Device Handler. The signon sequence will fail if the value is invalid.

The features supported by the above versions are summarised in the table below.

FEATURE	VERSION A	VERSION B	VERSION C
Card capture	Not supported	Supported	Supported
Message encryption in software	Supported	Not supported if hardware security is used	Not supported if hardware security is used
PIN PAD PIN Block encryption using hardware security module	Not supported	Supported	Supported
Signon data in a single message	Not supported	Not supported	Supported
Set date and time	Not supported	Not supported	Supported
PIN change transaction	Not supported	Not supported	Supported
Management of cash reduction	Not supported	Not supported	Supported
Current terminal totals	Not supported	Not supported	Supported
Dynamic key change	Not supported	Supported	Supported
MACing	Not supported	Not supported	Supported
Withdrawal Surcharging	Not supported	Not supported	Supported
Envelope Deposit	Supported	Supported	Supported
NDD	Not supported	Not supported	Supported

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USER S COMMENTS ON THE OLIVETTI 6000 DEVICE HANDLER USER S GUIDE

Please assist us in our future documentation plans by sending us your comments. We suggest that you use the form below, but please feel free to supplement it with more detailed comment as necessary.

NAME :

DATE :

COMPANY :

ADDRESS :

A. PERSONAL EXPERIENCE

1. How long have you been using the product?

0-3 months

☐

3-9 months

☐

Over 9 months

☐

2. In what capacity do you use the product?

Project Management

☐

Development

☐

Operations/Network

☐

Test

☐

B. THIS DOCUMENT

1. Have you had any difficulty with the product where you feel that this document ought to have helped but failed to do so? If yes, please give details:

2. In particular, we would appreciate your comments on the following aspects of the document:

		YES	NO
CONTENTS/INDEX	Is it helpful and is the level of detail sufficient?	☐	☐
STRUCTURE	Is the structure of the document satisfactory?	☐	☐
USEFULNESS	Do you often use this document in your day-to-day work?	☐	☐

3. Have you found the document useful? If so, please tell us what you particularly like about it:

4. Please give examples of anything that you have found to be unsatisfactory and suggest any improvements that could be made:

C. CURRENT DOCUMENTATION

- | | YES | NO |
|--|--------------------------|--------------------------|
| 1. Does the current range of manuals/user's guides cover the areas of system use on which you require information? | ☐ | ☐ |
| 2. If coverage is inadequate, please suggest what other areas could be covered: | | |

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